

Rope Access Practitioner and Supervisor

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P1. A Brief History

Gravity Training was established in 1999 and is an IWH, Services SETA, as well as MER Seta accredited Training Provider. In 2008 Gravity Training successfully complied with the South African Qualifications Authority (Act 58 of 1995) to provide quality outcomes-based training and assessment in accordance with the relevant Unit Standards as set forth by SETA.

Based on extensive experience gained, our accredited work at height courses, including fall arrest (tower climbing), rope access, rope rigging, climbing equipment inspection and other courses, are presented across South Africa, the African Continent and the International market.

Since the 1970's, companies realised that erecting scaffolding or hiring cranes in order to perform work at height was a very expensive and time consuming solution. Thus, the world of Work at Height was born. Since then the need for quality training in these fields steadily rose and is still doing so today.

This course, **Rope Access Practitioner and Supervisor**, will be focussing on the rope access skills, advanced techniques and legislative knowledge needed to be able to rescue a casualty out of any rope access position as well as supervising and managing a rope access team. This course is based on the Unit Standards: US229996, 229997 and 230001. This course includes all IRATA requirements as well, however, when presented in South Africa the course will also conform to these Unit Standards to cover techniques and some of the more advanced skills required to manoeuvre on ropes as well as perform rope access rescues.

P2. To the Learner

There are inherent risks involved in working at height and these risks are also ever-present, if not more so, during training. To minimise these risks, experience and knowledge must be gained in order to safely navigate the hazards as they present themselves. In order to gain the required knowledge, this manual has been created as an aid, **and only an aid, for trained and experienced** work at height technicians and facilitators. It is not designed to be used in isolation or without the input from a person trained in using it. However, at Gravity Training, all of the facilitators undergo rigorous training programmes in order to ensure that they are well equipped and as informed as possible to present this text in such a way that it is easy to understand, remember and implement on site. To ensure that this is indeed the outcome of this course, the onus lies upon the learner to dedicate him/herself for the duration of the course to understand and practise the techniques that will be demonstrated during the entirety of this course.

P3. General Site Safety Rules

When on site for training or work the following rules must be adhered to at all times:

1. Always wear seat belts when travelling in, or operating vehicles.
2. Always use suitable personal protective equipment, a safety harness and fall protection equipment when working at height (harnesses must be secured at all times when working at height).
3. Never carry out work on electrical equipment, circuits and gear if you are not qualified to do so.
4. Never work under the influence of substances (alcohol or drugs) which are illegal or in excess of legal levels or where this impairs your ability to perform tasks.
5. Never exceed speed limits or travel at speeds which are dangerous for the type of road, vehicle, or conditions.
6. Never use a hand held phone whilst driving and only make calls by pulling over or using hands-free devices when it is safe to do so.
7. Always wear helmets when travelling on, or operating a motorbike.
8. No passengers on the back of bakkies and trucks.
9. Never undertake any street or underground work activities unless competent to do so.



A. Fall Protection Plan

A. Fall Protection Plan

1. Introduction to Fall Protection Plan (For IRATA learners please refer to ICOP and TACS)

Before any work at height may commence, all the fall risks on site must be addressed and documented in a fall protection plan (FPP).

Construction Regulations (CR) of 2014, Section 10 (CR 10) sets out all the requirements for such a plan.

The purpose of CR 10 of 2014 is to ensure that a contractor:

- Identifies,
- Evaluates, and
- Mitigates

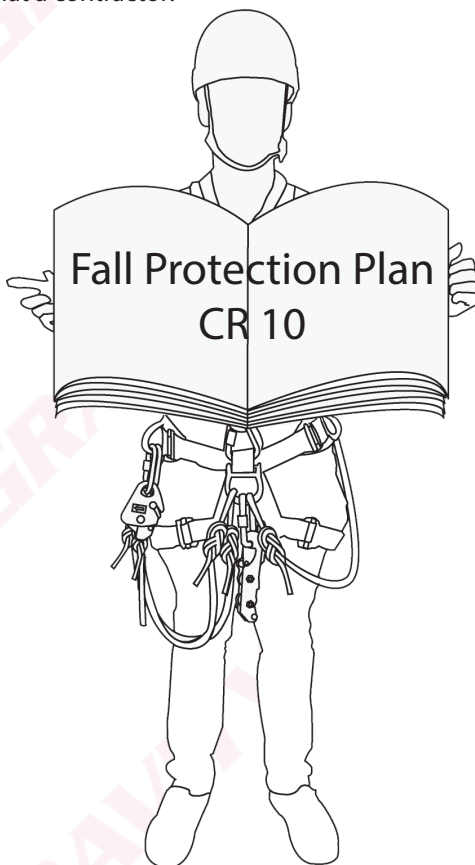
all prevalent fall risks and hazards on site.

According to CR 10(2) the Fall Protection Plan must include:

- CR 10(2)(a) - **Risk Assessment** - Forms the foundation of the plan.
- CR 10(2)(b) - **Medical Fitness** - All workers must be medically fit, and reviewed once a year.
- CR 10(2)(c) - **A Training Programme** - All workers must receive relevant training.
- CR 10(2)(d) - **Equipment Management** - All equipment must be "safe for use and fit for purpose". The equipment manager must be trained in the basic requirements of equipment management.
- CR 10(2)(e) - **Rescue Plan** - A trained rescuer must be able to rescue a worker from any situation.

A FPP must also be reviewed after any changes on site, or should the work scope change.

Note: All this must be done BEFORE work may start.



CR 18 - Rope Access Work

As per CR 18 - Rope Access Work - A competent person must be appointed as a Rope Access Supervisor in order to ensure that all rope access work is carried out safely.

An important part of this regulation is CR 18:

“(2) No contractor may use or allow the use of rope access work unless -

(a) the design and selection and use of the equipment and anchors comply with the safety standards ...

(b) he or she is in possession of the site specific fall protection plan.

(3) A contractor must ensure adequate measures are in place to allow rescue procedures to commence immediately.”

The reason this is included into this course is because it will be the responsibility of the Rope Access Supervisor to ensure that the above mentioned is achieved.

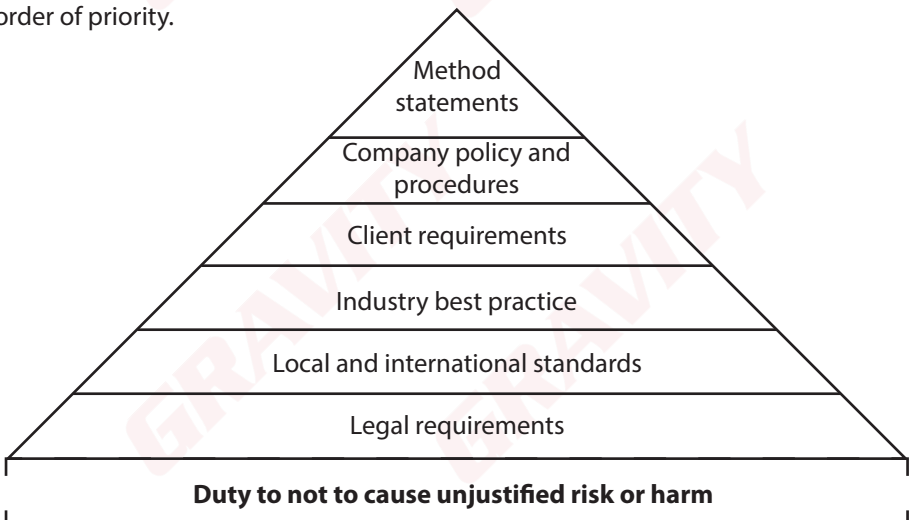
The Rope Access Supervisor is also responsible for preparing an operation plan/procedure and a risk assessment, as well as obtaining any and all required permits.

Section 24 of the OHS Act 85 of 1993 - Incident Reporting

Should a Site Supervisor be called upon to report any incident on site, section 24 of the OHS Act gives all the details of the procedure to be followed.

Hierarchy of Authority:

The following pyramid shows the various sources of requirements to be followed in order of priority.



2. Risk Assessment

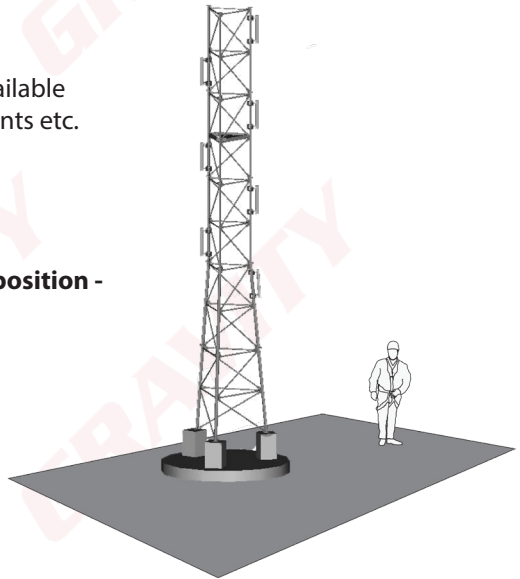
The most important part of the Fall Protection Plan is the **Risk Assessment**.

STEP 1 : IDENTIFY POSSIBLE RISKS

What to do when approaching a work site

BEFORE any work may start on a site the following must be identified:

- **Site requirements -**
 - What type of structure is available
 - What are the PPE requirements etc.
- **Access method requirements -**
 - Fall arrest
 - Rope access
 - Other
- **What is the required team composition -**
 - How many team members
 - Who must do what
 - Who will be who's buddy
- **Fall protection requirements -**
 - Work restraint
 - Fall arrest
 - Lifelines etc.
- **PPE requirements -**
 - Shock absorbing lanyards
 - Backup devices
 - Descending devices etc.
- **Scope of work**
- **Tools required for work**
- **Possible risks on site that may cause a fall**
 - e.g. bad weather, dangerous animals etc.



After this, the risks must be communicated to everyone on site by means of:

- Site induction
- Toolbox talk

All of the information on page 3 must be recorded in a “Operation Plan” similar to the one given below:

Team Details		Client	ABC Construction
Team Members	Position/Title	Location	Menlyn Mall
J. Smith	Level 3	Work site	Level 6 food court
A. Smith	Level 2	Client Contact	L. Nkosi
S. Smith	Level 1	Start Date	2016.01.01
M. Smith	Level 1	End Date	2016.01.07

Equipment Requirements (include a sketch of rigging arrangements)
4 x Rope access kits 2 x Straight drops 1 x Loop 2 x Drills with drill bits 40 x bolts with eye-nuts

Areas of concern
Dropping tools Unsupervised work Technicians getting stuck inside loop

Prepared by:	Signature:	Date:

With this information we can now prepare a Risk Assessment.

Below is an example of a risk assessment. Note that this is only an example and it is of utmost importance that the PRINCIPLES behind successful risk mitigation are understood and not just the format of the document.

When a risk is identified, it needs to be analysed and given a Risk Rating - it is given a rating twice - once before any control methods have been implemented and once afterwards. In order for us to consistently analyse our identified risk, and ensure that we can reproduce any rating, we need a “documented method” for this process as stated in CR 9(1)(c).

STEP 2 : PRE-MITIGATION RISK RATING

		HAZARD SEVERITY				
		Negligible	Slight	Moderate	High	Very High
LIKELIHOOD OF OCCURRENCE	Very Unlikely	LOW	LOW	LOW	LOW	LOW
	Unlikely	LOW	LOW	LOW	MEDIUM	MEDIUM
	Possible	LOW	LOW	MEDIUM	MEDIUM	HIGH
	Likely	LOW	MEDIUM	MEDIUM	HIGH	HIGH
	Very Likely	MEDIUM	MEDIUM	HIGH	HIGH	HIGH

No work is allowed to commence if there is a “Medium” or “High” risk present on site.

Thus, we have to mitigate all identified risks in order for them to receive a “Low” risk rating.

Hazard	Pre-mitigation Risk	Mitigation	Post-mitigation Risk
Unsafe use of rope access	High	• Ensure competent supervision. • Ensure all workers are competent for task to be performed. • Ensure all workers are medically fit. • Ensure all equipment is safe for use and fit for purpose.	Low

Again, the above is just an example of what a risk assessment MAY look like. When preparing one for a site, always remember to mitigate all risks thoroughly and ensure that mitigations used do not introduce new risks on site.

Step 3 (shown on the next page is a guide on how to mitigate risks in a responsible and manageable manner.

STEP 3 : RISK MITIGATION HIERARCHY

This is a process to help a company develop the correct procedures or methods to manage the risks in a reasonable and practical way.

STEP 1: Avoid the risk completely

Do not do the work; this will prevent the worker from being exposed to any risk.

STEP 2: Isolate the risk

Prevent workers from entering the area where there is a fall risk, **for example:** erect a solid barrier or handrail so that there is no risk of falling.

STEP 3: Engineer the risk away

Eliminate the need to access the fall risk area, **for example:** by locating or relocating items elsewhere if they require inspection and maintenance. Provide alternate means of access (which removes the fall risk), **for example:** walkways and telescopic handles or stairways instead of ladders etc.

STEP 4: Use fall prevention systems

Use suitable fall restraint systems and PPE that will prevent workers from reaching an edge where a fall may occur.

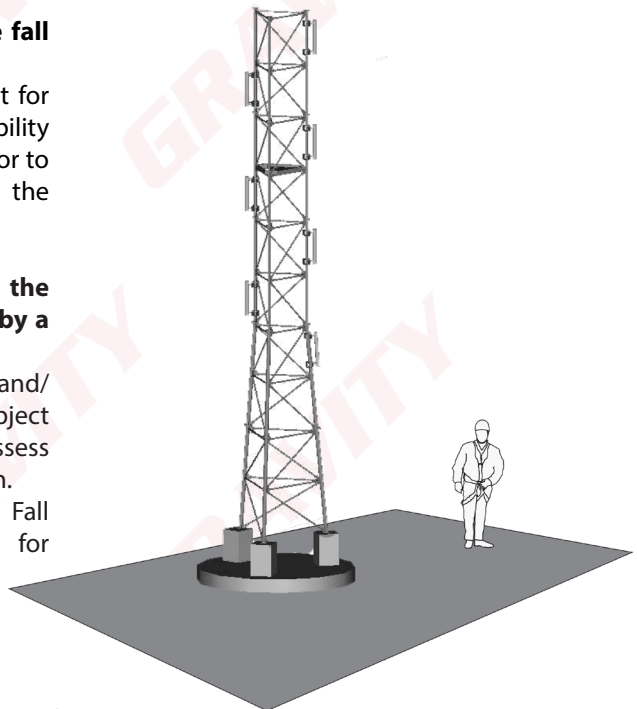
STEP 5: Select appropriate fall arrest system

Which is “safe for use and fit for purpose”. Ensure the availability of a competent site supervisor to identify access method and the correct rescue plan.

STEP 6: Stop work until the risks have been mitigated by a competent person

Consult the Site Supervisor and/or a work at height Subject Matter Expert (SME) and re-assess the work-site and rescue plan.

Alternatively, contact the Fall Protection Plan Developer for advice.



Step 6 can cause some controversy, so here are a few guidelines on when to stop work:

- **Adverse Weather Conditions:**

No work-at-height shall be done during any inclement weather conditions unless all risks have been mitigated adequately. Should the working situation change whilst working at height, all technicians must safely descend to safety and the Site Supervisor must report back to the Project Manager immediately. Examples of inclement weather conditions:

- Rain
- Lightning
- Extremely cold weather
- Extremely hot weather
- Sleet or snow
- High or gusty wind
- Poor visibility (fog or smog)



When working in windy conditions, the Beaufort Wind Force Scale can be used to determine the safety of the work-site. The Site Supervisor must ultimately take responsibility and work can be done if the risk assessment allows the working team to continue. Wind conditions must be continually monitored during the task as the conditions may change.

- **Environmental Factors:**

Apart from weather, any environmental factor that poses a danger to a worker must be addressed before work may begin. If this factor only becomes present after work has already begun, work has to be stopped until the risk has been mitigated. Examples of this include bees, snakes, thorn bushes, gasses, poor visibility, etc.

No work-at-height shall be done at night, unless adequate lighting is available on site, such as head-lamps or floodlights. Any hazards pertaining to lighting must be mitigated accordingly in the risk assessment.

- **Breach of Risk Assessment:**

If, at any time, any person on site, operates outside of the risk mitigations agreed upon during the toolbox talk, work must halt immediately and the person must either be removed from site (if for example it is a member of the public), or the toolbox talk must be repeated and explained again. This includes, but is not limited to:

- A person working without correct PPE.
- A person who is not medically fit.
- A person who has not been adequately trained etc.

STEP 4 : POST MITIGATION RISK RATING

After the mitigation, the risk must receive another “risk rating”.
Now they MUST be “**Low**”; if not, the process must be repeated.

		HAZARD SEVERITY				
		Negligible	Slight	Moderate	High	Very High
LIKELIHOOD OF OCCURRENCE	Very Unlikely	LOW	LOW	LOW	LOW	LOW
	Unlikely	LOW	LOW	LOW	MEDIUM	MEDIUM
	Possible	LOW	LOW	MEDIUM	MEDIUM	HIGH
	Likely	LOW	MEDIUM	MEDIUM	HIGH	HIGH
	Very Likely	MEDIUM	MEDIUM	HIGH	HIGH	HIGH

↓

Class Exercise 1

Complete the Risk Assessment below

Hazard	Pre-Mitigation Risk	Tick	Mitigation	Tick	Post-Mitigation Risk	Tick
Falling from Height	High		Correct PPE		High	
			Always Work Alone			
	Medium		Correct Rescue Procedures		Medium	
			Work in Teams of at Least Two (Buddy System)			
	Low		Diving Instruction		Low	
			Work at Height Training			

Risk Event	Risk Event Trigger	Possible sources for risk events	Risk Assessment			Hierarchy of risk controls	Risk controls to be implemented	Risk Assessment		
			Likelihood	Severity	Risk Rating			Likelihood	Severity	Risk Rating
Person falling	Work Platform related	Wrong type for task and competence								
		Unprotected edges								
		Open holes								
		Fragile surfaces								
	Access Method related	Over reaching from behind protection								
		Inadequate controls or supervision								
		Incorrect access method for task and area								
		Unsafe access methods/SOP								
		Inadequate controls or supervision								

Risk Event	Risk Event Trigger	Possible sources for risk events	Risk Assessment			Hierarchy of risk controls	Risk controls to be implemented	Risk Assessment		
			Likelihood:	Severity	Risk Rating			Likelihood:	Severity	Risk Rating
Rescues	Competence	Inadequate supervision for rescue								
		Do not have the correct competence and SOP for scenario								
		Not enough man power for scenario								
		Cannot perform first aid during rescue								
	Rescue Kit	Insufficient kit to perform rescue								
		Inadequate equipment to perform rescue								
	Refresher Training	Unfamiliar rescue kit and SOP								
		Unsure of the SOP to follow								
		New HSE risk not taken into account								

B. Duties of a Rope Access Supervisor

1. Plan and Prepare

As a Rope Access Supervisor, it is of great importance to adequately plan and prepare for all work well in advance in order to ensure that all scenarios and risks are covered in accordance with the OHS Act 85 of 1993 as well as ISO 22846-1 - Rope Access - Fundamental Principles and ISO 22846-2 - Rope Access - Code of Practice .

Consider the following:

- **PPPPP** - Proper Planning Prevents Poor Performance. Planning your work properly beforehand will ensure that the job is completed with minimal complications or delays.
- **Time management** - It is easier and quicker to do something right the first time rather than doing it over and over again. Taking your time when establishing a work site, and ensuring that everything needed is in place, will save a lot of time by preventing the need to re-rig or change tack.
- **Team selection** - As a Rope Access Supervisor, it is your duty to analyse the site and determine the difficulty of the proposed tasks, and choose your team members accordingly. Only level 1's, level 1's and 2's or only level 2's for example.
- **PPE/equipment selection** - Ensure that all workers on site are trained to use PPE and equipment to be used as determined by the risk assessment for the task to be performed.
- **Selection of the rigging set-up** - As a Rope Access Supervisor, you will be responsible for deciding how the ropes must be rigged when establishing a work site.

2. Delegation of Work

A level 3 Rope Access Supervisor may be needed in more than one place at the same time - a challenging feat - however, a level 3 may delegate supervision to a level 2 in the following circumstances.

- The Level 2 is competent to supervise - The level 3 is to analyse the site and to ensure that the scope of work is not beyond the capabilities of a level 2.
- The Level 3 has inspected, analysed and approved all risk mitigations and preparations.
- The Level 3 has determined that there will be no need for complex manoeuvres or rescues.

3. Team Psychology and Attitude

Apart from the responsibilities upon the shoulders of the a level 3 to supervise and ensure that all rope access work is carried out safely, he/she must also determine whether workers are fit to work at height with regards to their psychological state. There are many factors that may influence the mental state of a person such as:

- Loss of a loved one - The tragic and traumatic experience of losing a loved one may lead to a person being unable to focus or concentrate.
- Post Traumatic Stress Disorder - any person who has experienced a traumatic experience (such as a fall from height for example) may suffer from PTSD. This can lead to unforeseeable changes in personality and can lead to unpredictable actions.
- Fear of heights - Many people might not display or even know that they are afraid of heights when exposed to low level heights, however it may be a completely different story should the person find him/herself higher up.
- New technician - Due to inexperience, a technician may be stressed or afraid and may make mistakes due to lack of concentration. In such a case, unlike the others, the Supervisor can ensure that they do not work alone and are constantly under the supervision of a more experienced, higher qualified team member.

Apart from the examples mentioned above, debt, depression, anxiety, stress or fatigue, to name a few, can also cause changes in the psychological make-up of a person. Supervisors should always be aware of the psychological state of their team members in order to ensure that they are fit for work and able to work safely.

The Level 3 Supervisor does have the right to remove any worker from site should he/she feel that the person jeopardises the safety of either him/herself, a team member or a third party.

The same applies to the attitude of team members. A bad attitude towards safety, risk assessment, procedures, following instructions or general work ethic can just as easily endanger workers as medical or psychological factors. Any worker displaying an attitude detrimental to the group as a whole must be dealt with accordingly by the Site Supervisor.

4. Selection of Work Procedure

As stated earlier, there exists a range of techniques on how to stay safe at height, fall arrest, work restraint, rope access etc. It is the duty of the Rope Access Supervisor to select the suitable method, or write one should none exist, to ensure safe work at height.

5. Just Safety Culture

Just Safety Culture refers to the 3 duties imposed on each individual in order for them to work safely.

These 3 duties are:

- A duty to not cause harm or risk - conduct oneself in such a fashion as to not harm oneself or others, or increase any risk to harm.
- A duty to follow procedural rule - always follow procedures as they are laid out, it is each person's duty to familiarise themselves with any and all relevant procedures. Ignorance is no excuse.
- A duty to produce an outcome - when all is said and done, one must still complete the task that you were set to do.

6. Consequences of Neglecting Supervisor Duties

It is your duty not to cause harm. Negligent or inadequate supervision will immediately endanger all workers on site. This can lead to incorrect rigging, incorrect use of equipment or deviation from safe work procedures.

Should a person injure him/herself on site, an investigation will be launched. Should it then be found that this was due to inadequate supervision, there may be legal repercussions as per Section 37 of the OHS Act (Occupational Health and Safety Act 85 of 1993).

7. Obtaining Work Permits

No work is allowed to commence without a work permit and official clearance. This to ensure that all relevant persons are aware of workers on site and are able to manage any risks and tasks accordingly. For example, should work be required in confined space, a gas test needs to be performed before work may start. Thus, a permit and clearance will ensure that this has indeed been done and will ensure that no worker is exposed to a hazardous environment. It is the duty of the client to ensure that the site is safe and to give permission for the work to commence, however, the Site Supervisor must ensure that all relevant documentation is in place.

8. Completion and Signing of Logbooks

As a Rope Access Supervisor, it will be your responsibility to inspect and sign the logbooks of all technicians on your team. Ensure that they are completed correctly and honestly as all hours claimed here will be used to prove competency to advance to the next level of rope access.

Ensure that the workers only claim hours in which they were actively busy with rope access work. Time spent on planning and preparation may also be claimed.

Date: _____	Work Site: _____
Time Arrived on Site: _____	Time Leave Site: _____
Hours Worked: _____	Employing Company: _____
Job Details: _____	

Supervisors Signature: _____	
Print Name: _____	

Date: _____	Work Site: _____
Time Arrived on Site: _____	Time Leave Site: _____
Hours Worked: _____	Employing Company: _____
Job Details: _____	

Supervisors Signature: _____	
Print Name: _____	

RUNNING TOTAL HOURS WORKED _____

When signing a level 2 rope access technicians’ logbook, add whether he/she was acting as a supervisor or not under job details, as this will allow him/her to prove supervision experience should he/she advance to level 3.

C. Work at Height Basics

1. Shock Load

The force experienced by an object when its fall comes to a stop, is referred to as a shock load. It is this force that causes damage to equipment, injury and even death.

It should be quite obvious then that avoiding this force is of the highest priority in the work at height industry. All systems and equipment used during work at height must be subjected to analysis in order to determine what the effects might be on the worker and/or equipment. Should this analysis show that the worker or equipment may suffer any harm, the system or equipment must be changed.

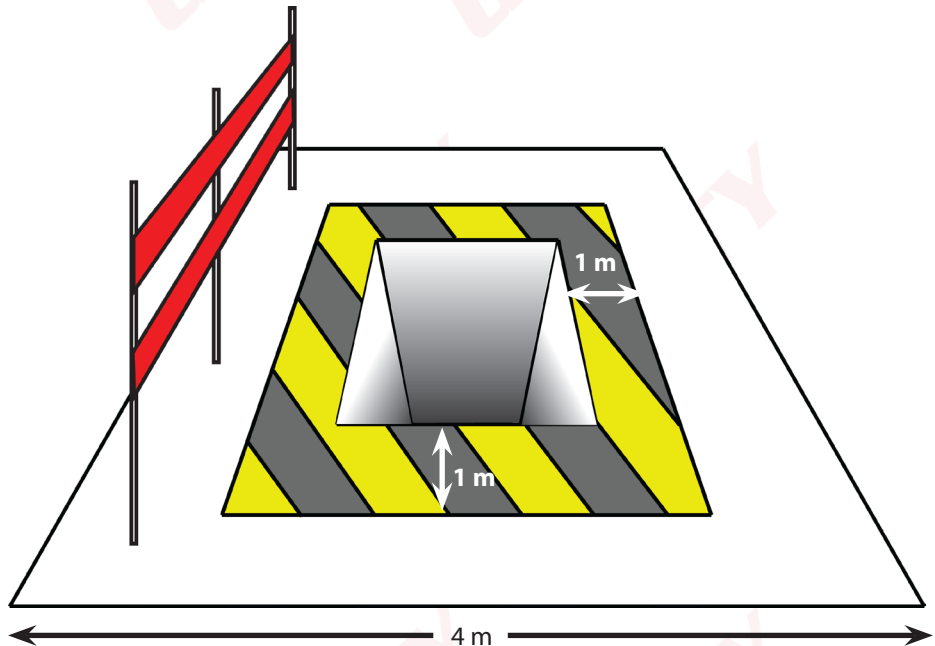


It is the duty of the Rope Access Supervisor, to ensure that no worker or system is in the position where a shock load would cause damage or harm during work or rescue.

2. Exclusion Zones and Barricading

The easiest way to prevent a fall/shock load is by preventing workers from reaching a fall risk position. This can be done by one of three methods:

- **Exclusion zones:** a zone or space where people are **not allowed** to walk or work, for example, a 1 m space around a hole in a roof. This is an example of **passive** crowd control.
- **Barricading:** a physical barrier **preventing** people from reaching a fall risk position. This is an example of **passive** crowd control.
- **Access Control Officer:** a member of the work team that controls access and stop people from entering the work area. This is an example of **active** crowd control.

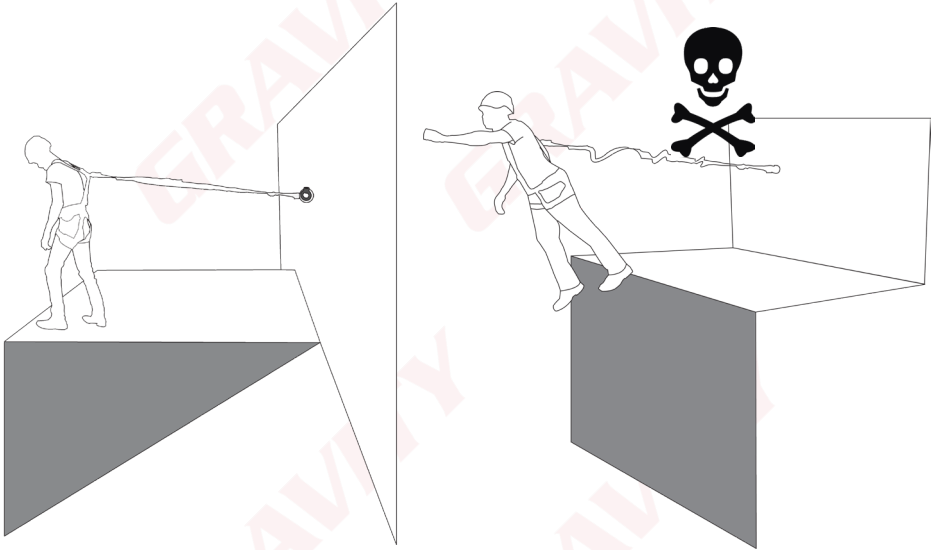


Exclusion zones can either be created at the top of the work site to prevent the public/other workers from falling or dropping items to a lower level, or at ground level to prevent dropped items or persons falling onto the public/other workers.

Should both of the above mentioned methods not be possible for whatever reason, a person may be made safe by use of a work restraint system, or fall arrest system.

3. Work Restraint

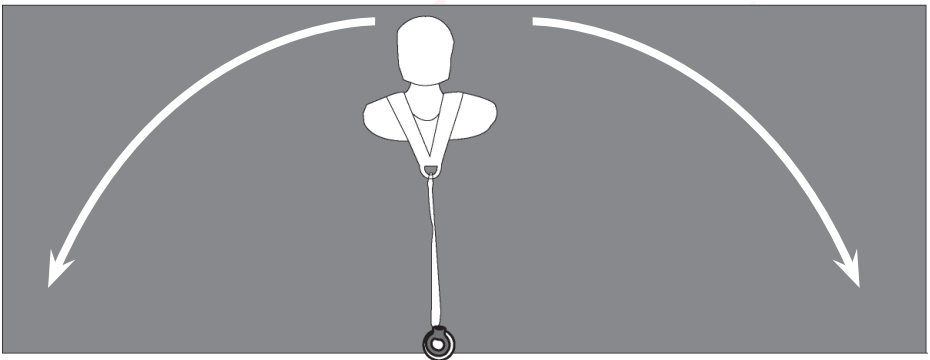
Definition: A system where we physically tie a worker to the structure, thus preventing him/her from reaching the edge, thus preventing a fall.



Advantages: Cheap, effective, little training required and minimal equipment required, no shock load, simple rescues.

Disadvantages: Technician's movement is limited.

Remember to ensure that there is no way that you can reach any of the edges surrounding the work area.

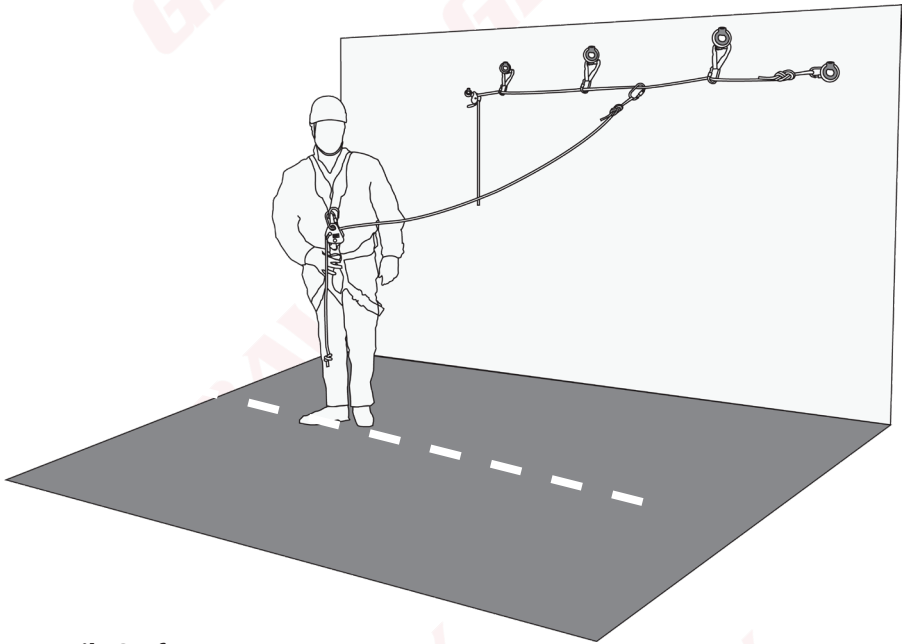


4. Rooftop Work Restraint System

One can combine the horizontal lifeline with a work restraint system to create a system ideal for rooftop work.

This allows a technician to move left to right and the technician can manually manage the distance to reach the required working area.

Special care should be taken when building the rooftop work restraint system to prevent the person from reaching the edge of the work area.



5. Fragile Surfaces

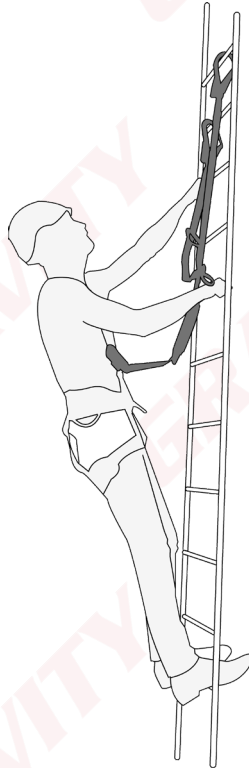
An important part of rooftop work is to remember that there is always a risk that the roof has been neglected or damaged over time, leading to the surface becoming fragile, which can cause a worker to fall through the roof. To prevent this, ensure that the roof is inspected by the relevant responsible person and that you are permitted to work upon it. This must be covered in your risk assessment before work starts.

6. Fall Arrest

Definition: a system used to suspend a person safely AFTER a fall has occurred, and doing all this in such a way as to prevent injury as far as possible.

Fall Arrest does NOT prevent a fall like Work Restraint. It only prevents a person from getting hurt or hitting the ground due to a fall.

A basic Fall Arrest system consists of a physical connection between the worker and the structure, but with some sort of shock absorbing component. Otherwise, it will be a piece of equipment designed in such a way that the worker's fall is arrested so quickly that there is no time to generate a notable shock load.



Advantages: Cheap, effective, and good freedom of movement.

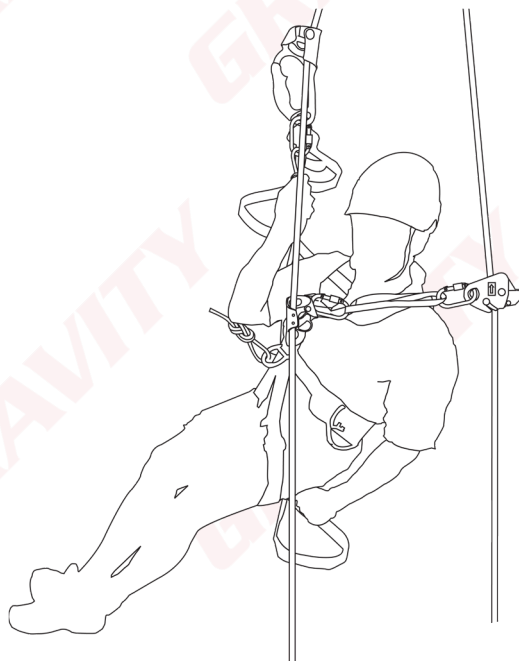
Disadvantages: Requires more training than work restraint, chance of shock load, rescues not as simple as with work restraint.

Now, for us to work safely, and to be able to use either a work restraint or fall arrest system, we need our Personal Protective Equipment or PPE.

7. Rope Access

Definition: a form of work positioning, using a series of connection points and ropes to enable access to difficult to reach locations where other safer methods are not possible.

Much like Fall Arrest, Rope Access does NOT prevent a fall like Work Restraint does. It only prevents a person from getting hurt or hitting the ground due to a fall. Rope Access also employs a double rope system that reduces the likelihood of a ground fall greatly.



2-Point Principle

A rope access technician must, at all times, have two points of connection, i.e. one point on each of the two ropes. Failure to do so will put the technician in grave danger of a ground fall.

Advantages: Access to difficult positions, cheaper than alternative methods (scaffold, MEWPS, etc.).

Disadvantages: Requires more training.

Now, for us to work safely, and to be able to use either a work restraint or rope access system, we need our Personal Protective Equipment or PPE.

8. Fall Factors

To determine the severity of a fall, we need to calculate the "Fall Factor".

The formula used to determine a Fall Factor:

$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

There are three main Fall Factors:

Fall Factor 0

Fall Factor 1

Fall Factor 2

Fall Factor 0: ✓✓

Fall Arrest

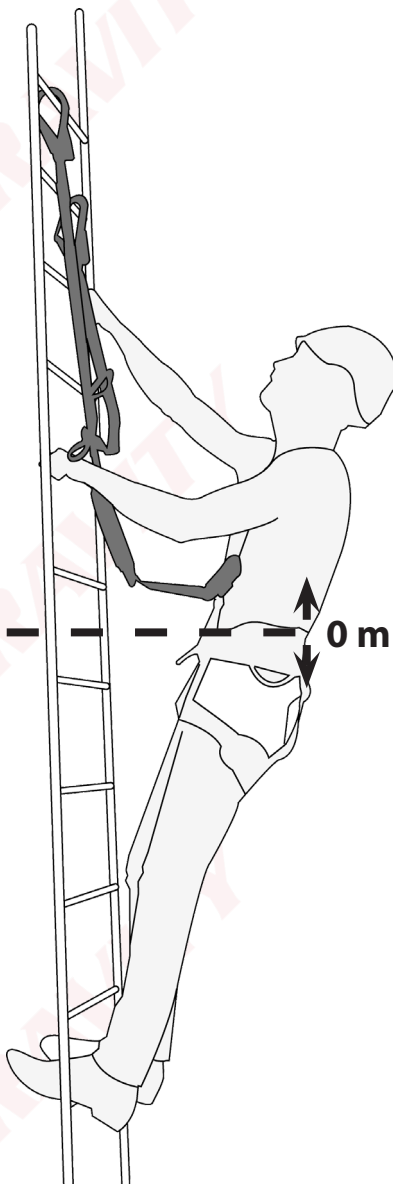
This is the ideal position to be in. Hooking up as high as possible above the head will result in:

- Almost no falling distance.
- Negligible shock load.
- No deployment of the shock absorber.
- No injury.

$$\text{Fall Factor} = \frac{\text{distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{0 \text{ m}}{2 \text{ m}}$$

$$\text{Fall Factor} = 0$$



Rope Access

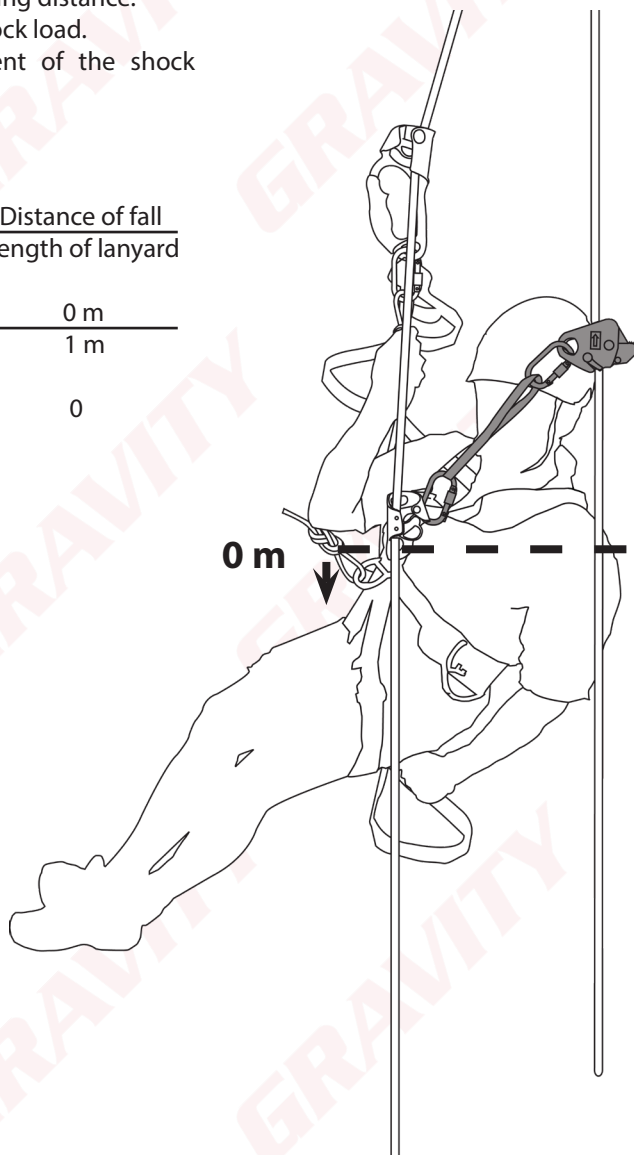
This is the ideal position to be in. Hooking up as high as possible above the head will result in:

- Almost no falling distance.
- Negligible shock load.
- No deployment of the shock absorber.
- No injury.

$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{0 \text{ m}}{1 \text{ m}}$$

$$\text{Fall Factor} = 0$$



Fall Factor 1: ✓

Fall Arrest

This position, hooking up at the same level as the chest is not as safe as a Fall Factor 0, but is still suitable.

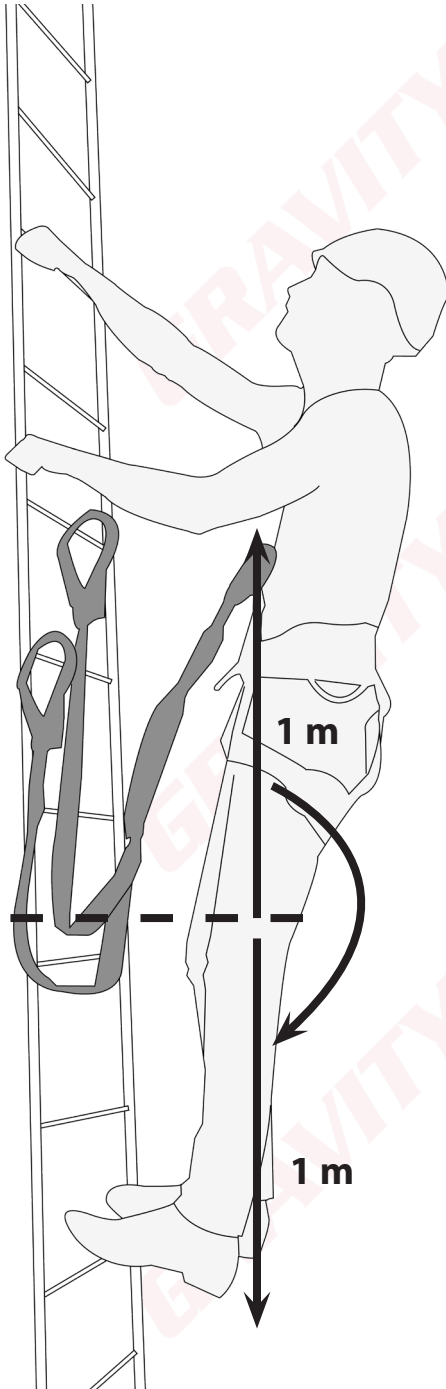
Falling from this position will result in:

- Medium falling distance.
- Slight shock load.
- Partial deployment of the shock absorber.
- Possible slight injury (bruises, whiplash etc.).

$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{2 \text{ m}}{2 \text{ m}}$$

$$\text{Fall Factor} = 1$$



Rope Access

This position, hooking up at the same level as the chest, is not as safe as a Fall Factor 0, but is still acceptable.

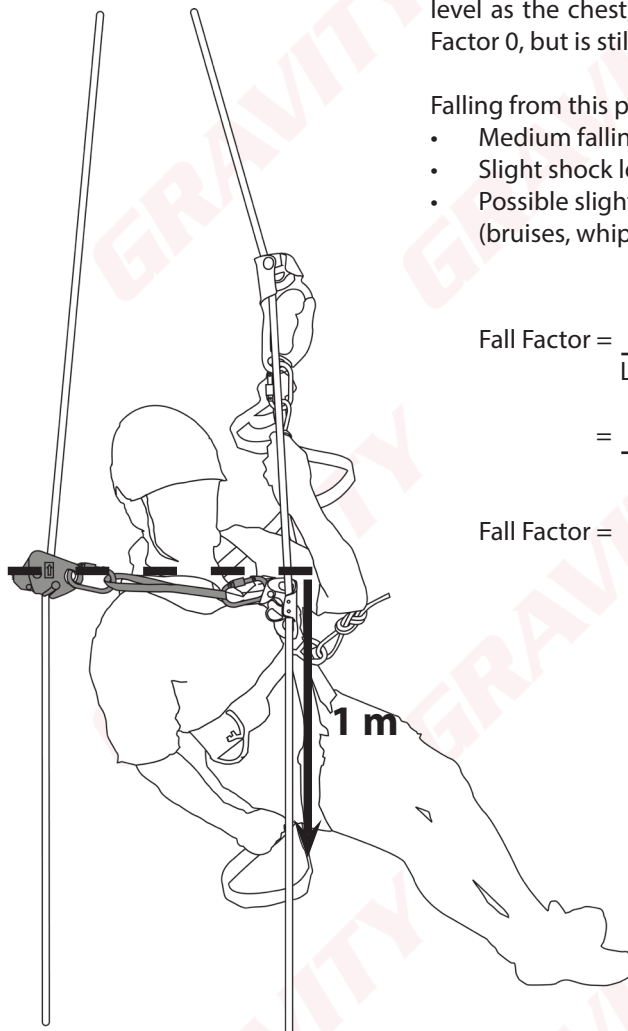
Falling from this position will result in:

- Medium falling distance.
- Slight shock load.
- Possible slight injury (bruises, whiplash etc.).

$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{1 \text{ m}}{1 \text{ m}}$$

$$\text{Fall Factor} = 1$$



Fall Factor 2:

Fall Arrest

This is the most dangerous position to be in. No worker should ever be allowed to work from this position; if there is no higher connection point, a different fall protection system must be used.

Falling from this position will result in:

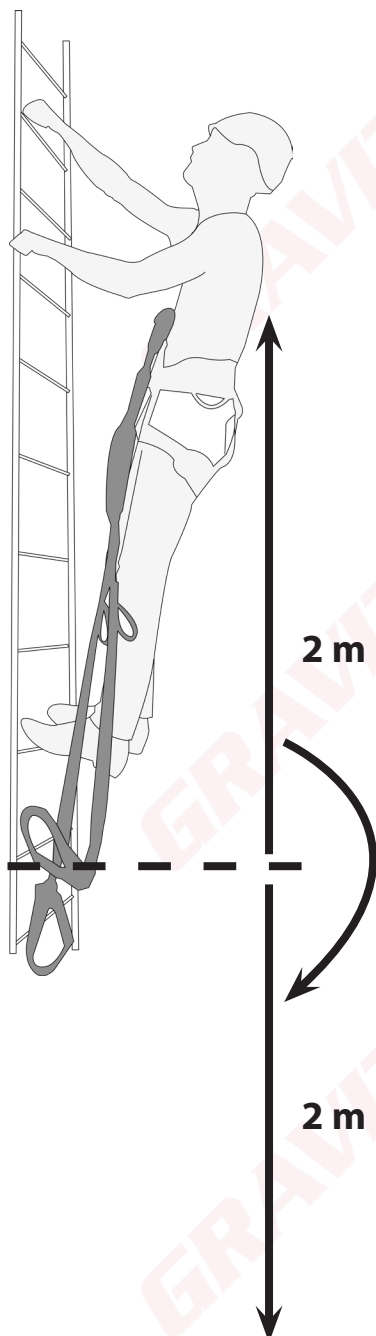
- A large falling distance.
- Maximum shock load.
- Full deployment of the shock absorber.
- Definite injury (broken bones, back injury etc.).



$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{4 \text{ m}}{2 \text{ m}}$$

$$\text{Fall Factor} = 2$$



Rope Access

This is the most dangerous position to be in. No worker must ever be allowed to work from this position:

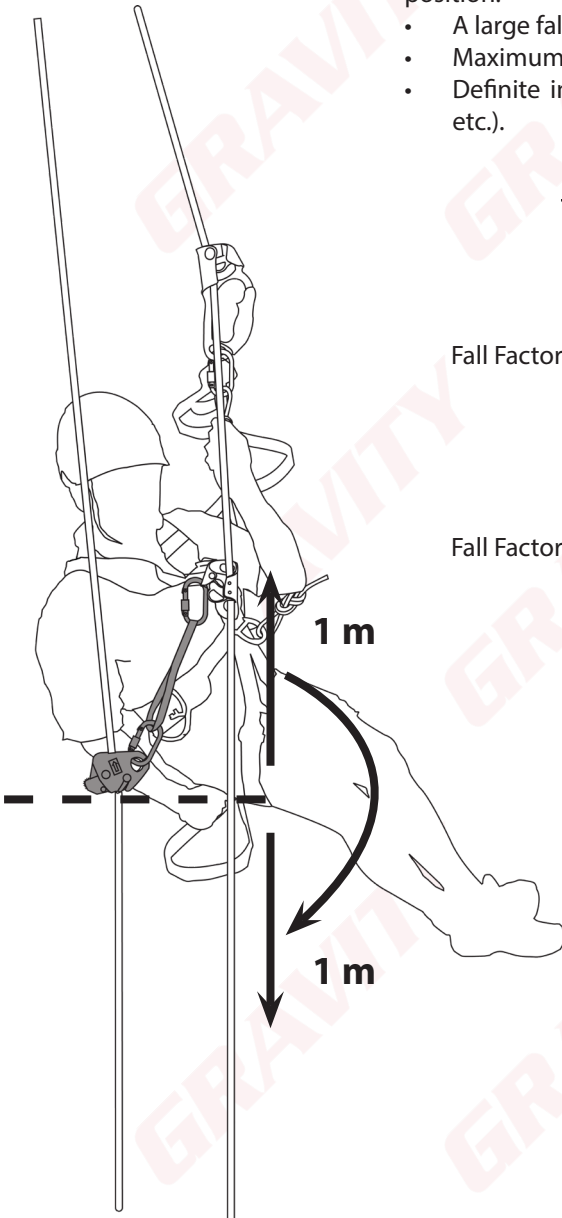
- A large falling distance.
- Maximum shock load.
- Definite injury (broken bones, back injury etc.).



$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{2 \text{ m}}{1 \text{ m}}$$

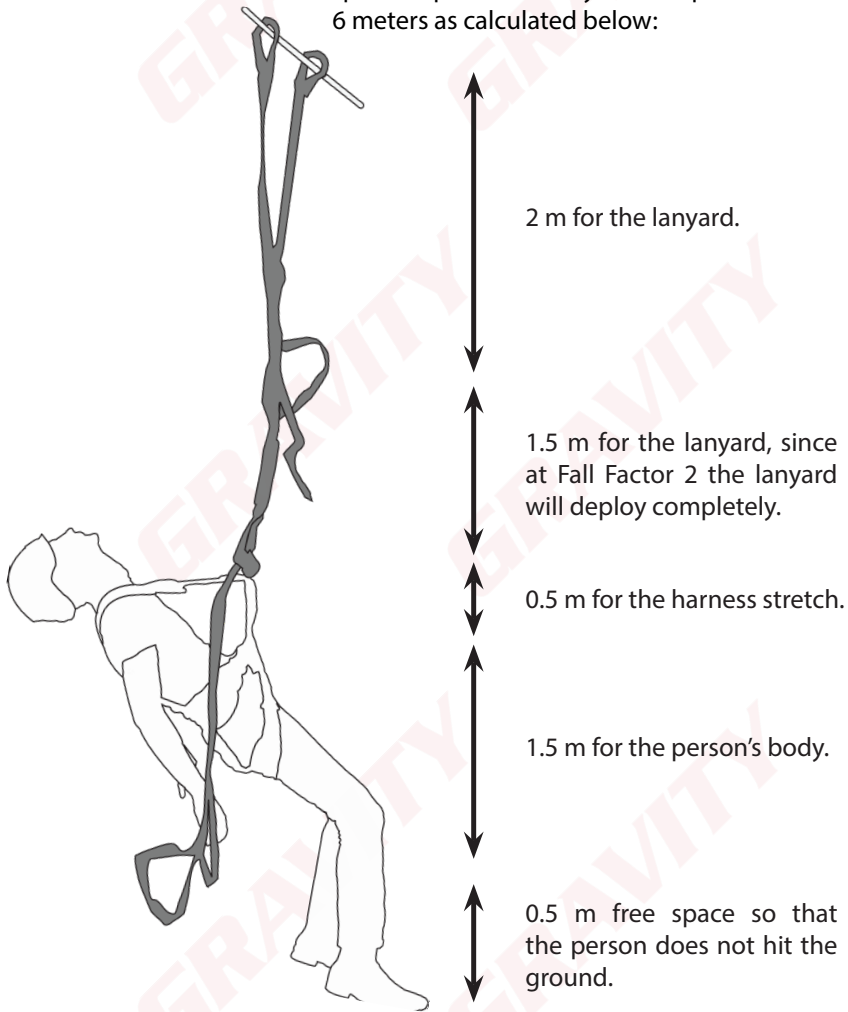
$$\text{Fall Factor} = 2$$



2. Minimum Free Space

Minimum free space (MFS) is the open space a technician would need below him to safely arrest a person's fall without hitting the ground or a structure. Reducing the fall distance or Fall Factor would reduce the required MFS.

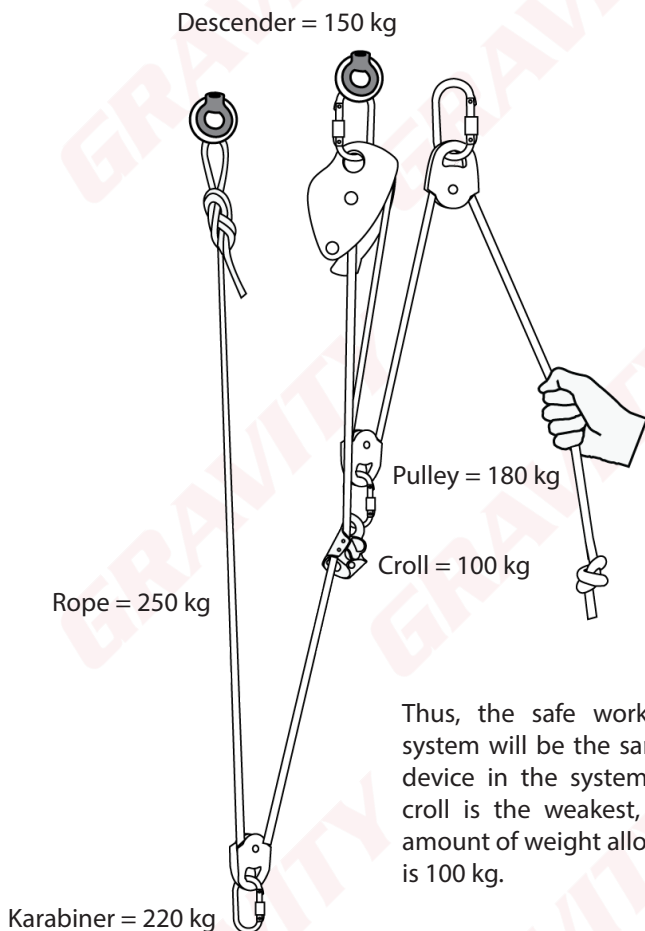
For example, if a person working in a Fall Factor 2 with a 2 m lanyard falls from a structure, the minimum free space required to safely arrest a person's fall would be 6 meters as calculated below:



THUS, $2\text{ m} + 1.5\text{ m} + 0.5\text{ m} + 1.5\text{ m} + 0.5\text{ m} = 6\text{ m}$

10. The Weakest Link

When combining multiple pieces of equipment, always consider the “weakest link in the chain” as this will be the **MAXIMUM AMOUNT OF WEIGHT** you may put on the system.



Thus, the safe working load for this system will be the same as the weakest device in the system. In this case, the croll is the weakest, so the maximum amount of weight allowed on the system is 100 kg.

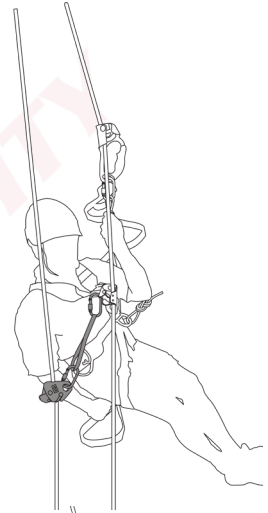
It is apparent then, that it is impractical to rig an entire 6 to 1 system if it only allows a worker to lift 100 kg. The options of equipment should always be considered in terms of safe working loads. A Rescuescender, in the case above, would allow a worker to lift up to 150 kg, for example.

Class Exercise 2

Identify the Fall Factor of the worker in the picture to the right.

- A) Fall Factor 1.
- B) Fall Factor 2.
- C) Fall Factor 0.
- D) Fall Factor 3.

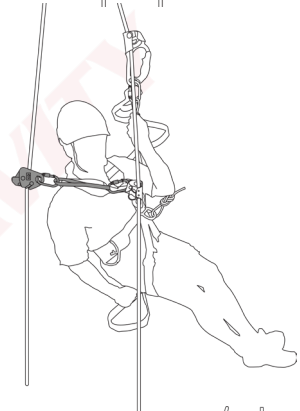
Answer: (A) (B) (C) (D)



Identify the Fall Factor of the worker in the picture to the right.

- A) Fall Factor 1.
- B) Fall Factor 2.
- C) Fall Factor 0.
- D) Fall Factor 3.

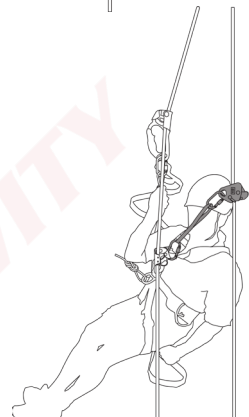
Answer: (A) (B) (C) (D)



Identify the Fall Factor of the worker in the picture to the right.

- A) Fall Factor 1.
- B) Fall Factor 2.
- C) Fall Factor 0.
- D) Fall Factor 3.

Answer: (A) (B) (C) (D)



D. Climbing Equipment

1. Requirements of Climbing Equipment

According to ISO 22846 -2, the minimum knowledge of equipment is as follows:

- **Selection of equipment** - must be safe for use and fit for purpose, and conform to the standards relevant to the intended use. **ISO 22846-2(7.4)** gives the specific requirements for selection of equipment.
- **Limitations of individual items of equipment or systems** - equipment must only be used in accordance with manufacturer's user instructions.
- **Pre-use inspection** - supervisor must ensure that all workers on site conduct a full and thorough inspection of all equipment to ensure that all equipment is safe for use and fit for purpose.
- **Detailed (Formal) Inspection** - According to ISO 22846 7.7.2.2 equipment must undergo a regular detailed inspection at intervals not exceeding 6 months. However, equipment that is frequently used must be inspected at shorter intervals. Refer to the company risk assessment to determine whether equipment should be inspected even more frequently.
- **Storage, cleaning and basic maintenance** - all equipment must be stored, cleaned and maintained in accordance with manufacturer's requirements.
- **Identification of defects** - as a Rope Access Supervisor, you will be qualified to conduct 6-monthly equipment inspections, therefore, it is essential that you familiarise yourself with all inspection criteria of any and all equipment that will be used on site.
- **Discard procedures** - should any defects be identified during the 6-monthly inspections or the pre-use inspections, a discard procedure must be in place, or one should be written.

As you can see all of the above mentioned information hinges on the fact that the Rope Access Supervisor is familiar with both the individual certification standards of all equipment, as well as in-depth knowledge of the ISO 22846 1 & 2 standards.

As a Rope Access Supervisor, you will be responsible for all of the above, and to ensure that all team members comply to the ISO requirements.

PPE INSPECTION SHEET as Used By:								
Rope Grab (EN 353-2)								
Inspected by:			Date of Inspection:			Date of Next Inspection:		
No.	ID/Serial No.	Body	Wear & Tear	Working Action	Bolts & Rivets	Spring Action	Rust	Pass/Reject
1								
2								
PPE INSPECTION SHEET as used by:								
Shock Absorber (EN355)								
Inspected by:			Date of Inspection:			Date of Next Inspection:		
No.	ID/Serial No.	Date of Manufacture	Discolouration	Stitching	Deployment	Wear & Tear	Connectors	Pass/Reject
1								
2								

E. Bolting

As a level 3 Rope Access Supervisor, you will be qualified and required to place and test anchor points. You may delegate this duty to a less qualified rope access technician, but he/she may only place or test anchors under the direct supervision of a level 3.

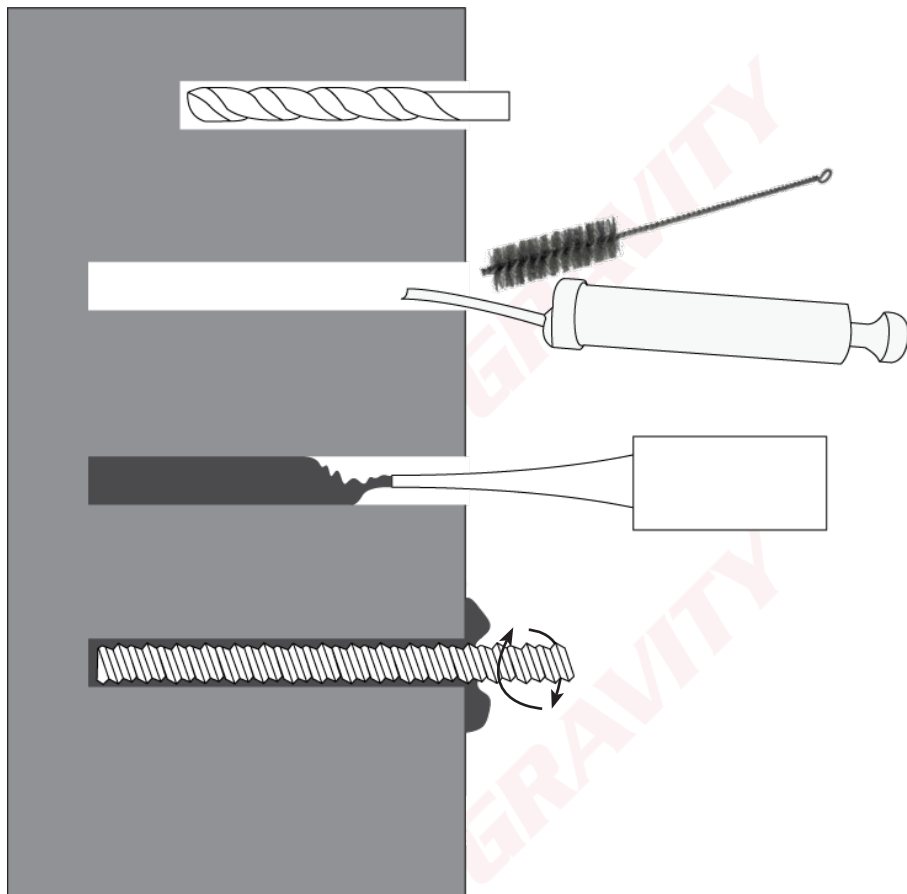
1. Requirements of Bolts

- When bolting into masonry, bolts shall be of stainless steel.
- When bolting into steel, the bolt material shall match that of the structure.
- All bolts shall have a stainless steel hanger or eye with load rating as recommended by the manufacturer.
- All bolts shall have a minimum diameter of 10 mm.
- Manufacturer's requirements shall be followed at all times.

2. Bolt Placement

- Should any doubt arise regarding the placement of bolts, ISO 22846-1/2 states that the person placing the bolts should refer to the OEM requirements. Otherwise, SANS 10333 (although it is no longer in use) gives detailed instructions for the safe placements of anchor points.
- Bolts shall be placed by a qualified person (a level 3 Rope Access Technician) or a technician working under his/her direct supervision.
- When bolting into a parapet wall, the bolt shall be placed at a minimum of 500 mm from the top of the wall.
- No bolts shall be placed less than 300 mm from the edge, and expansion bolts shall not be placed closer than 500 mm from any edge.
- When using chemical anchors, the chemical shall be of an injectable type.
- Holes drilled for chemical anchors shall be properly cleaned using a brush and blowing device. Care shall be taken not to allow any liquids to enter the hole. This includes saliva, sweat etc., as this can affect the chemical reaction.
- The depth of the drilled hole shall be a minimum of 100 mm, or deeper if so specified by the manufacturer.
- After placement, the hole shall be sealed against the ingress of water.
- Expansion bolts shall not be placed in bricks or weak substrates. These are only recommended for use in reinforced concrete.
- Should technical/complex bolting be required, contact an expert such as Gravity Training to assist and advise.





1. Drill the hole, at least 100 mm deep.
2. Clean the hole, brush first and then blow, until the dust is cleared out of the hole.
3. Fill the hole with chemical, starting from the bottom of the hole and taking care not to create bubbles. If installing a express anchor, insert anchor and hit it forcefully with a hammer to ensure the anchor is secure.
4. Insert the threaded rod, turning it as you do so to ensure that the chemical covers all of the thread of the rod.
5. Allow the chemical to cure.
6. Test the bolt.
7. Install an eye nut or hanger.

3. Testing of Bolts

In the case of chemical anchors, according to EN 795 / SANS 50795 and ISO 22846-1 & 2, after a bolt has been placed and the chemical has cured, it needs to be tested.

Planned anchor points are tested directly after placement, and then annually thereafter.

- **First test: 10 kN for 3 minutes**
- **Annual test: 6 kN for 30 seconds**

The standard recommends that the annual test is reduced to 6 kN for 30 seconds.

Typically planned anchor points are rated to 22 kN BS and our first test only affirms a SWL of 100 kg. Thus, it is good practice to always place 2 independent anchors in order to share the load, resulting in a 200 kg SWL for the entire system.



Factors that influence the strength of placed anchor points.	
Masonry	• Condition of bricks (cracks, age etc.) • Type of brick (holed, thin, baked etc.)
Steel	• Condition of steel (rust, cracks, bends etc.) • Type of steel (must match type of steel used for anchor)
Concrete	• Condition of concrete (cracks, age, strength)
Rock	• Condition of rock (cracks, strength)

F. Rigging

Safety Factors

DIFFERENCES BETWEEN LIFTING EQUIPMENT, AND ROPE ACCESS & FALL ARREST EQUIPMENT:

Lifting Equipment & Safety Factors:

- Lifting equipment will always have a WLL (Working Load Limit) that indicates the maximum load that can be lifted with the equipment.
- The rating will be in TON or KG.
- Lifting equipment also has a safety rating, for example 6:1, which implies that a 1 ton webbing sling will break at 6 ton. This will be indicated on the equipment.
For Example:
 - o Webbing Slings 6:1
 - o Wire Slings 6:1
 - o Chain Slings 5:1
 - o Lifting Shackles 6:1
- Lifting slings are colour coded according to strength and should not be confused with rope access equipment which is not colour coded.

Rope Access & Fall Arrest Equipment, and Safety Factors:

- Rope Access and Fall Arrest equipment will have a S.W.L (Safe Working Load) on the equipment that will indicate the amount of weight (in KN; 1 KN = 100 kg) that can be applied to the equipment safely.
- However the B.S (Breaking Strength) of equipment indicates the amount of weight the equipment can hold before it will break.
- In Rope Access and Fall Arrest equipment the relation between the S.W.L and the B.S will be the safety rating (not indicated on the equipment) and will always be 10:1.

Summary:

The big difference between lifting equipment and Rope Access / Fall Arrest equipment is that lifting equipment will always have a WLL (Working Load Limit) that means the maximum load that can safely be lifted with the equipment. The rating will be in KG or TON. Rope Access and Fall Arrest equipment have a breaking load rating. The rating will always be in KN; this will indicate the amount of force required in order for the equipment to break. Dividing this rating by a safety factor of 10, we arrive at the Safe Working Load.

1. Rigging of Work Ropes

All work ropes must be rigged in such a way that there is a main line (working line) and a backup line (safety line). Each line is to be anchored independently, with its own anchor point and connector. It is advised to rig in such a way that, should one anchor fail, there is a redundancy. In other words, each rope is to be rigged with two anchor points. In order to save on equipment, one can rig so that two ropes share two independent anchor points.

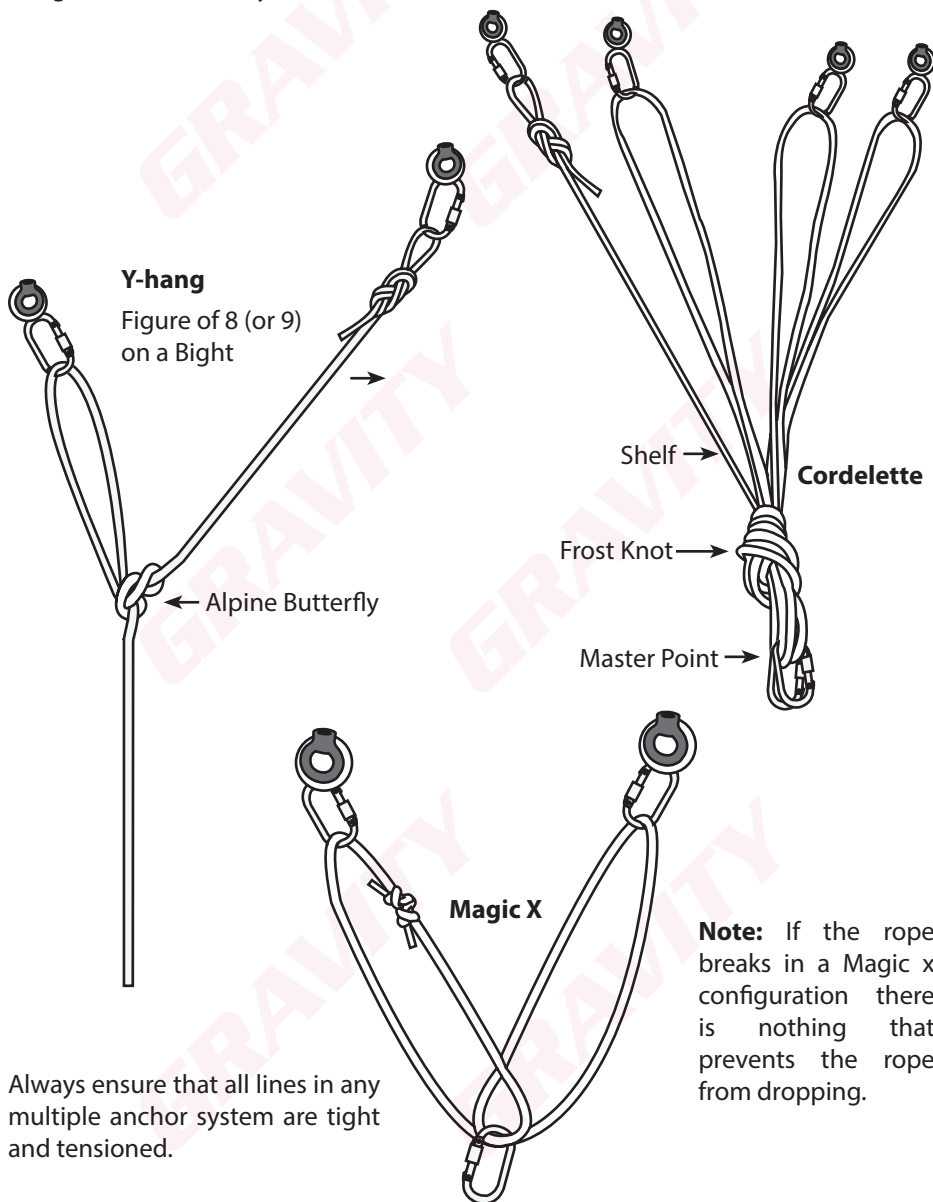
Main line, with Double Loop Figure 8 knot, anchored to two separate anchor points.

Backup line, with Double Loop Figure 8 knot, anchored to two separate anchor points.

Rigging two ropes with two independent anchor points with redundancy.

2. Rigging with Multiple Anchor Points

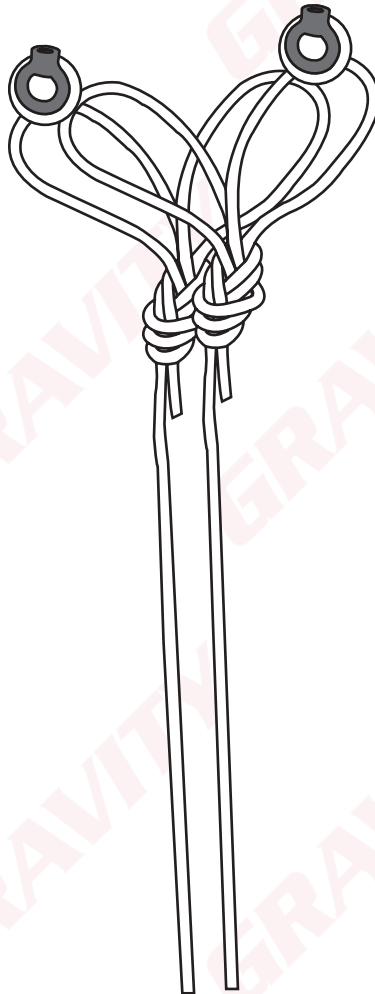
Typically, we use a Double Loop Figure 8 to rig work ropes, however, sometimes the position of anchor points do not allow this knot to be used, or it is desired to rig with extra redundancy. This also helps with sharing the load evenly. In this case, we need to rig with a different system.



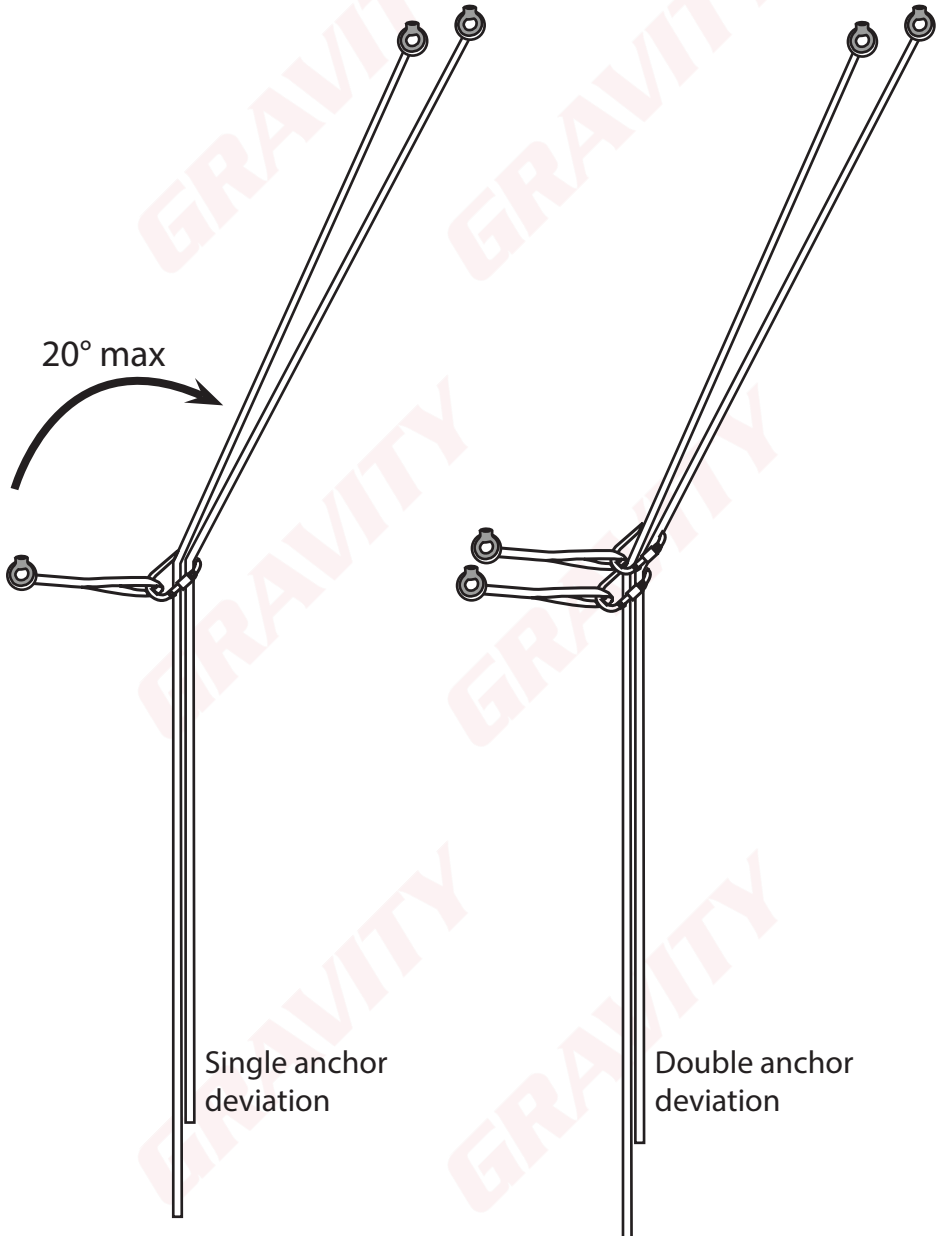
3. Selecting the Correct Rigging Set-up

When should one rig which type of rope set-up? Below follows a simple non-exhaustive guide.

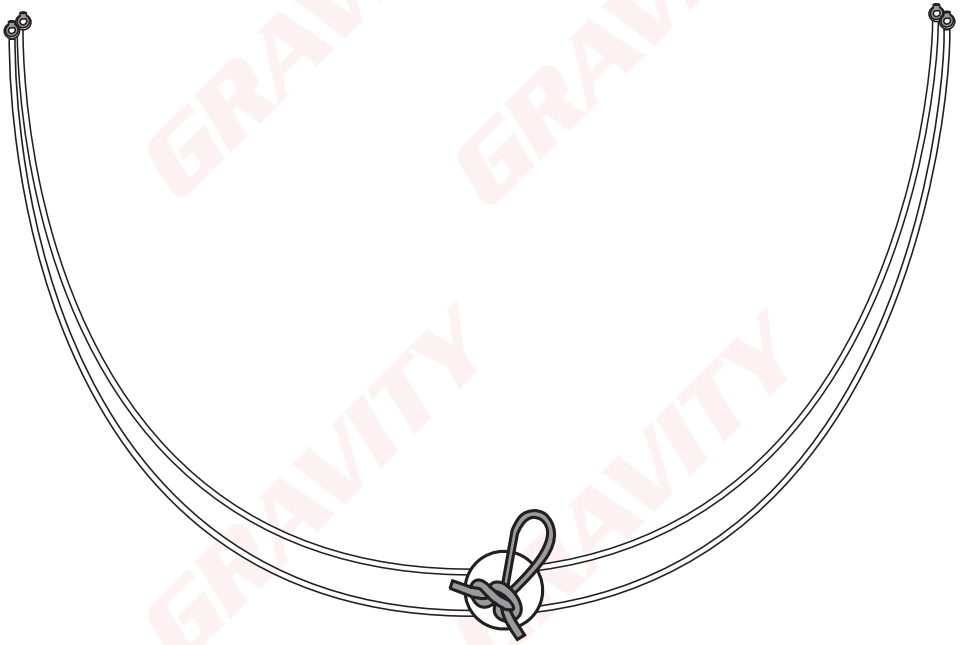
- **Straight drop:** This is the most commonly used set-up. Consider if this is suitable by looking to see where the ropes would hang, if there is any chance of the ropes becoming a hazard to other people on site, or perhaps there might be machinery of some kind that may damage the ropes.



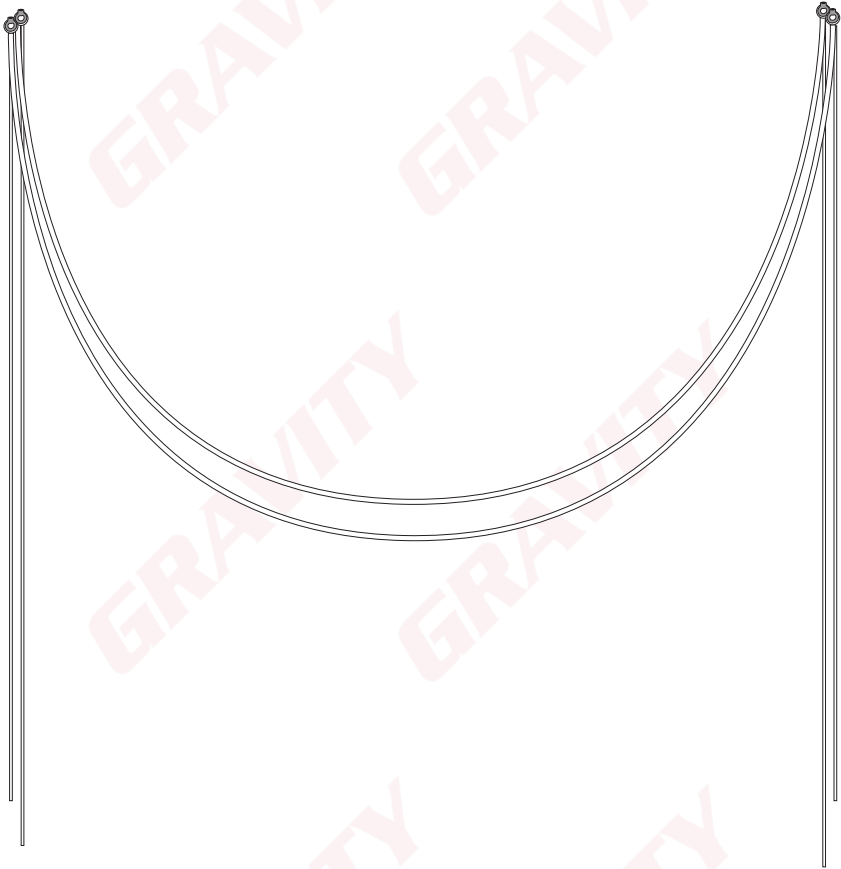
- **Deviation:** Should it, for some reason, not be possible for a worker to stand directly underneath the anchor points used for the rigging of the ropes, deviate the rope through an additional anchor point. This allows the worker to circumnavigate any potential hazards that prevents use of a straight drop.



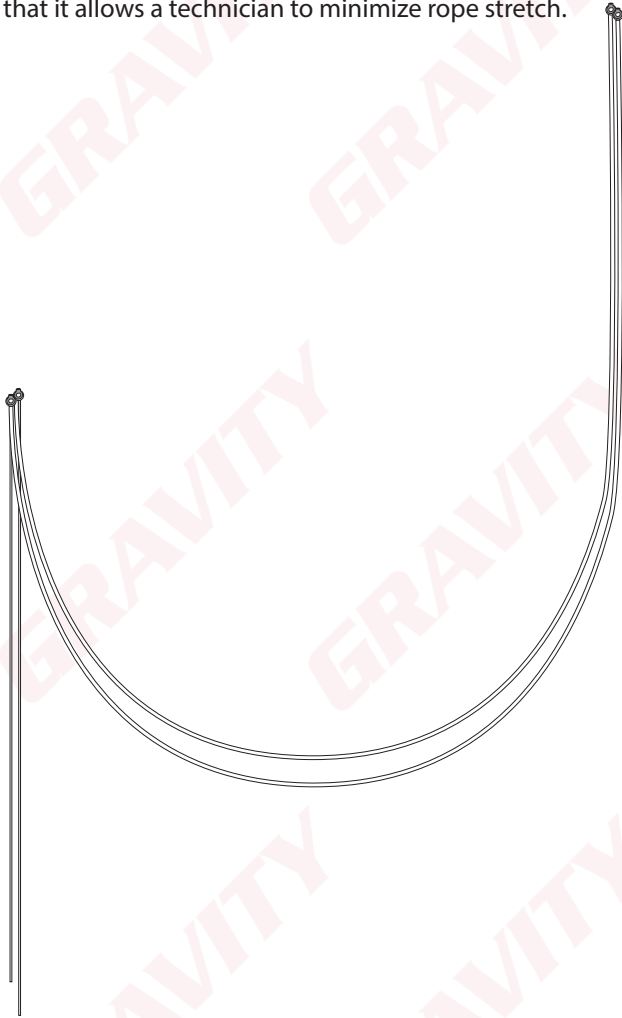
- **Long rope transfers:** This consists of two straight drops connected with a knot in the middle and allows the worker to move horizontally as well as vertically.



- **Loop:** Used for the same purpose as a long rope transfer, however a loop is used should there be a hazard underneath the ropes which may cause damage to the ropes or injury to the worker. Loops can also be rigged with different lengths of straight drops on either side according to site requirements.



- **Re-belay/re-anchors:** Can be used the same as a loop but with a straight drop on only one side of the loop. However, more commonly, with a re-belay the anchor point for the far side of the loop is much higher than the anchor point for the straight drop. This is useful when work must be done under and over an overhang or when a deviation cannot be used. Another useful purpose of a re-belay is that it allows a technician to minimize rope stretch.

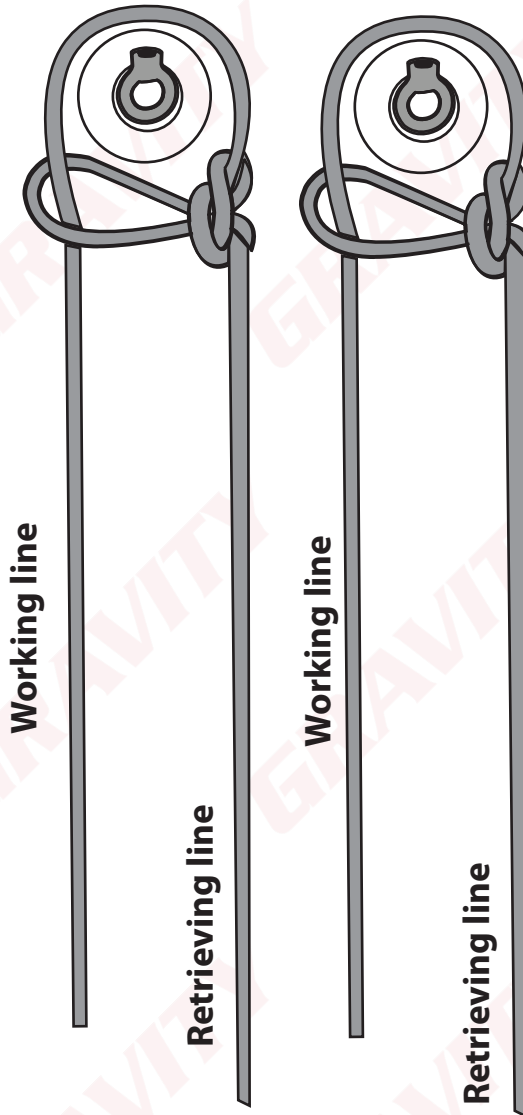


4. Rigging Over Sharp Edges

When sharp edges can't be avoided, there are 3 alternatives.

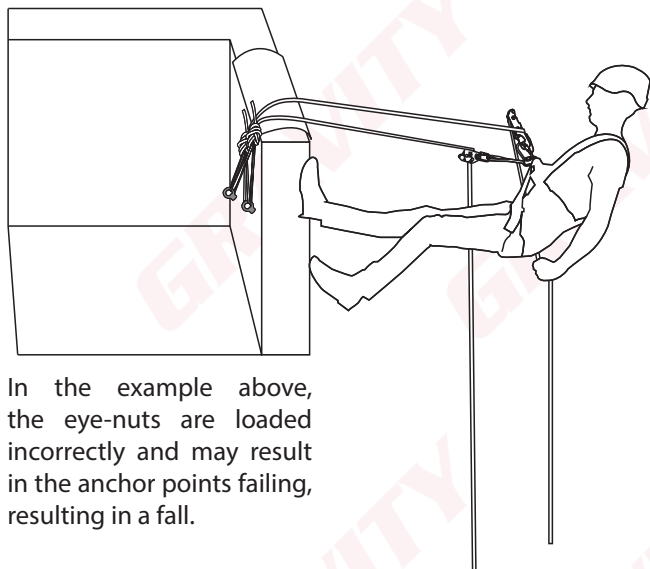
1. Modify the edge - grind or sand the edge smooth.
2. Layer the edge - install edge protection.
3. Change the system - rig the ropes in such a way that they do not touch the edge.

5. Retrievable Anchor



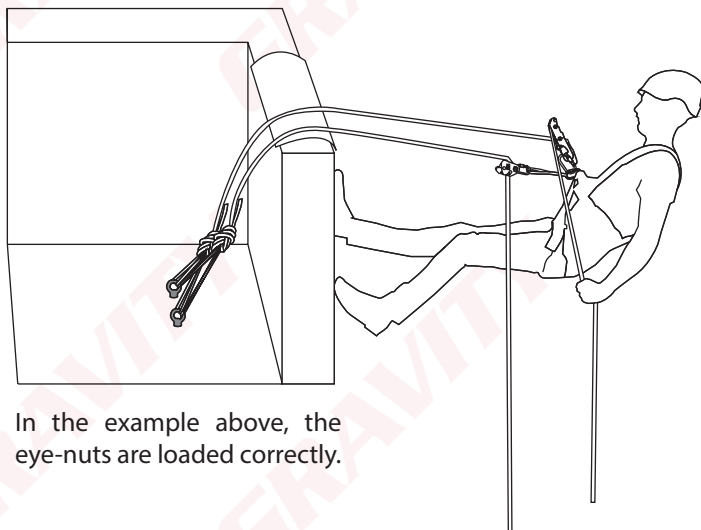
If the structure allows it, karabiners may be added between the Alpine Butterfly and the working line to minimise the soft-on-soft friction.

DO NOT rig ropes so that anchor points are loaded incorrectly.



In the example above, the eye-nuts are loaded incorrectly and may result in the anchor points failing, resulting in a fall.

Ensure that chosen anchor points are suited for the intended purpose.



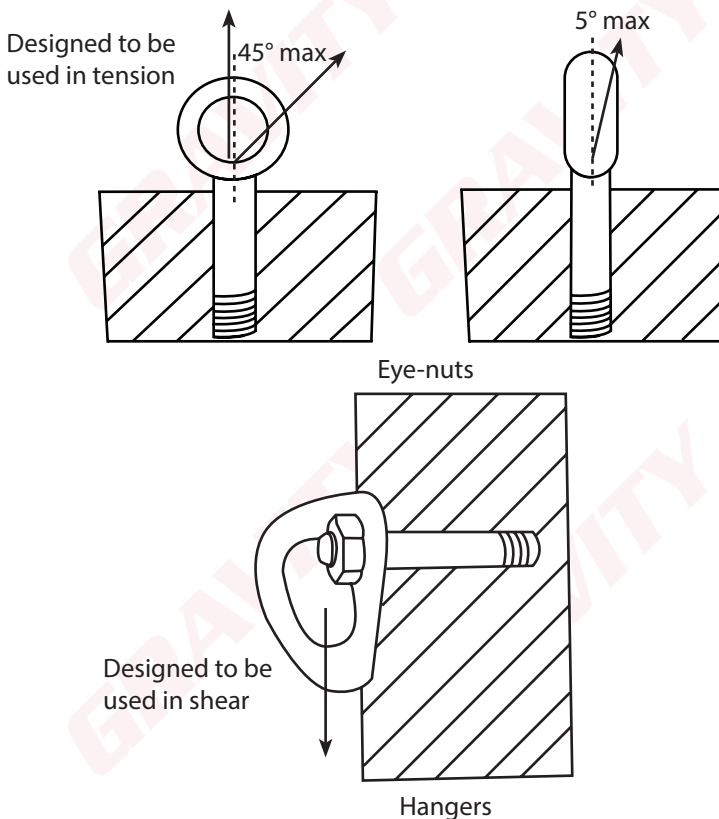
In the example above, the eye-nuts are loaded correctly.

6. Rigging and Slings Loads

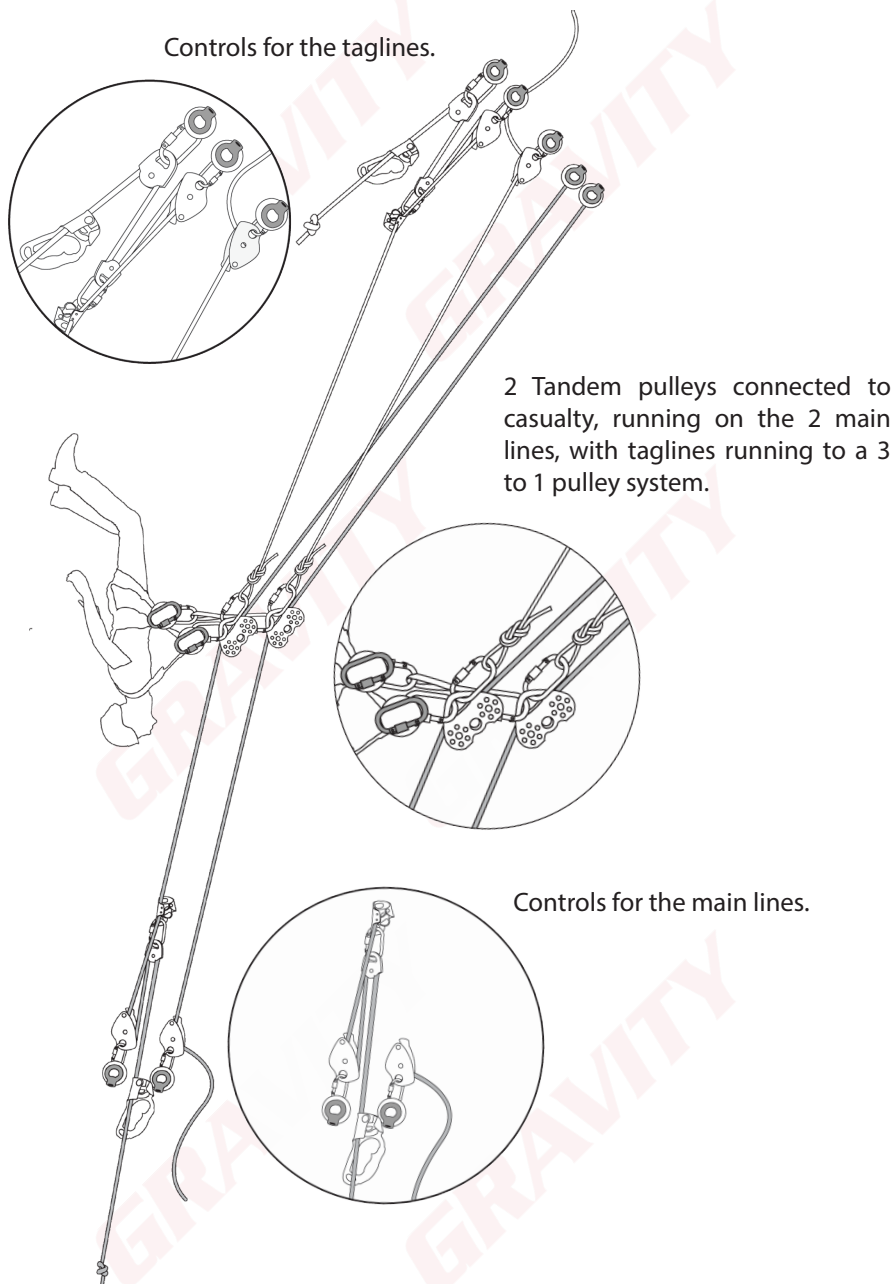
When preparing to lift any load, remember the following:

Principles of rigging -

- Always complete a risk assessment and lifting plan before work starts.
- Barricade any area where there is a risk of people coming close to the load.
- Always conduct a pre-lift.
- Never walk underneath a load.
- Always consider the centre of gravity of the load to ensure that it is as stable in the air as it would have been on the ground.
- Ensure that the equipment used to sling the load does not damage the load and vice versa.
- Always maintain the safe working loads and working load limits.
- When lifting loads over 40 kg, or when transporting a person, always ensure that the load or person is connected to two points.
- Always ensure that the anchor points are loaded correctly.

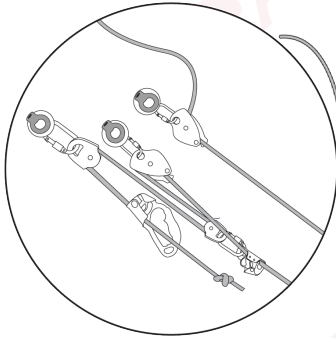


7. Rigging Diagonal Cableway



8. Rigging Horizontal Cableway

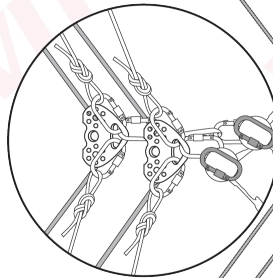
Controls for the main lines.



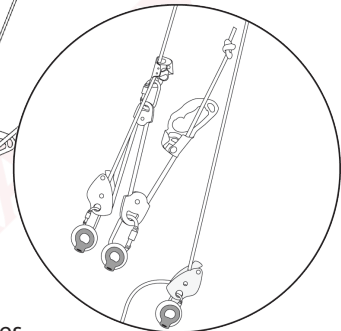
Controls for the taglines.

Note: only one 3 to 1 pulley system is needed here as the one can be switched from taglines to main lines as needed.

2 tandem pulleys connected to casualty, running on the 2 main lines, with 2 taglines running to 2 separate control systems.

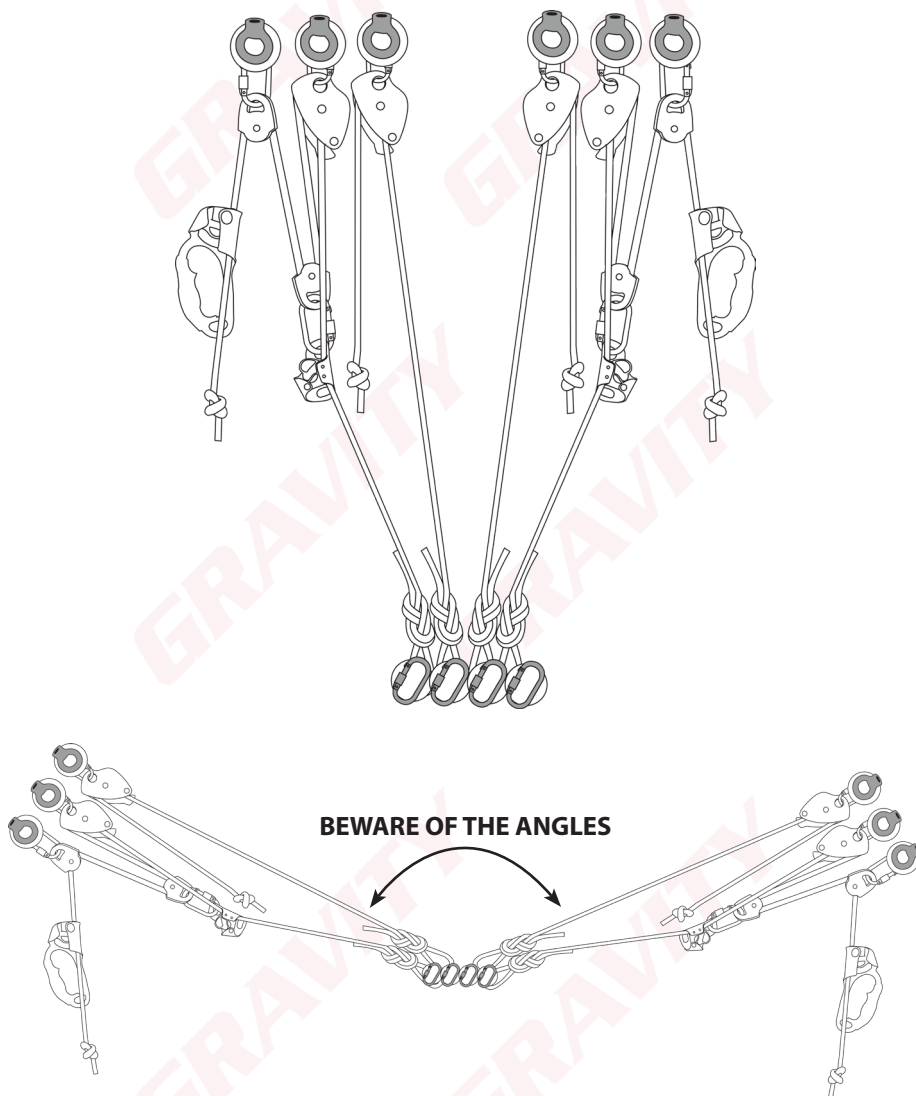


Controls for the taglines.



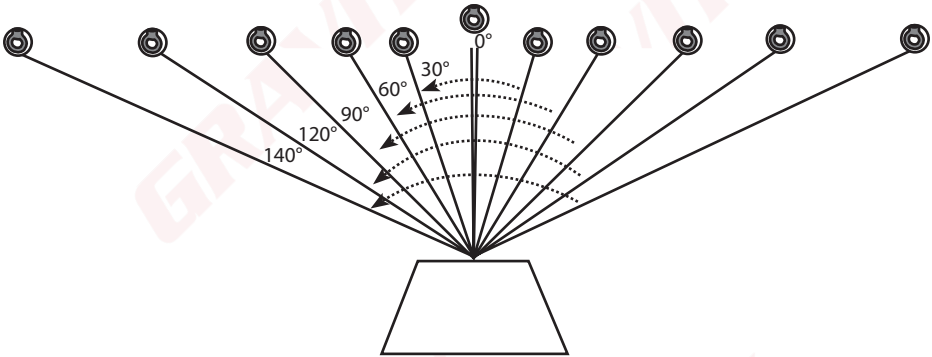
9. Rigging a Cross-haul

A cross-haul is a very simple, yet effective method of transporting any load horizontally as well as vertically. It consists of two 3 to 1 pulley systems connected independently to the load.



10. Angles and Forces

When rigging, the angle between anchor points determines the force experienced by each anchor. Below is a breakdown of the angles and their corresponding forces.

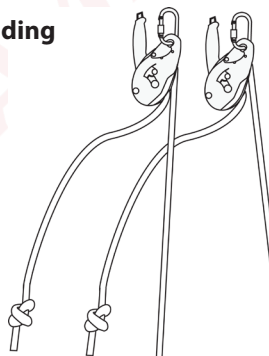


Angle	Force experienced by each anchor
0°	50% of load
30°	52% of load
60°	57% of load
90° (recommended)	70% of load
120° (maximum with restrictions)	100% of load
140° (dangerous)	150% of load
160° (very dangerous)	300% of load

11. Rigging for Rescue

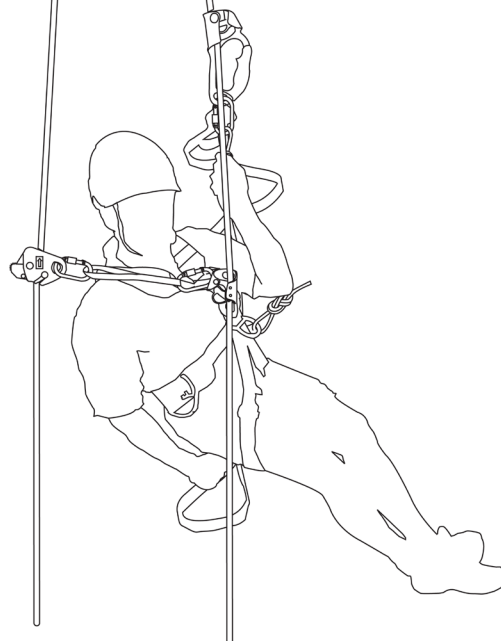
Prevention is better than cure. Thus, always consider to rig your work ropes in such a way as to make rescue as easy and as simple as possible. Below are two methods of rigging for rescue; one for lowering, and one for lifting.

Rig for Rescue: Descending



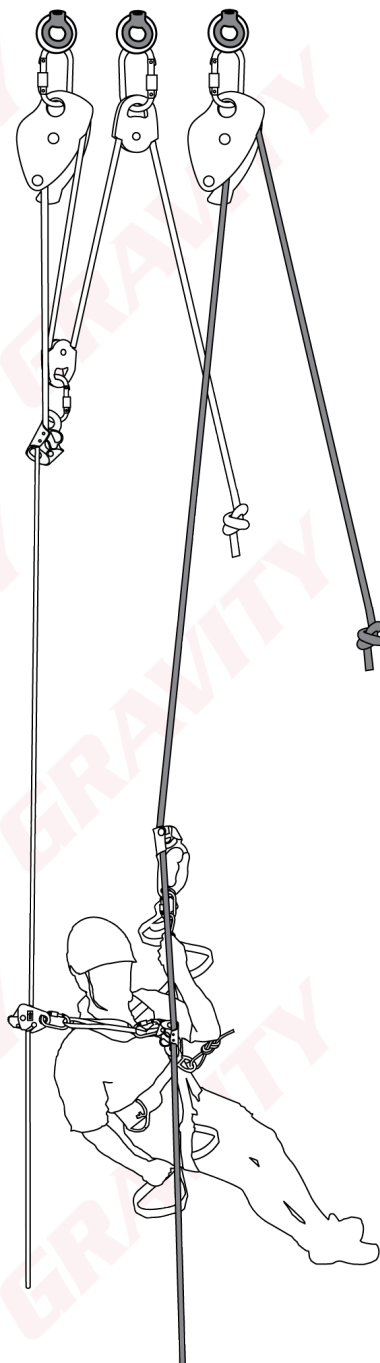
The main line and backup line is rigged through two descending devices. This allows the rescuer to simply lower the worker in need of rescue to the ground.

The length of the slack ropes need to be at least the same as the distance from the anchor points to the point of safety.



Rig for Rescue: Ascending

The main line and backup line is rigged through two descending devices of which one is equipped with a 3 to 1 pulley system. This allows the rescuer to lift the worker in need of rescue to the top of the line, resetting the pulley system when needed.



G. Pulley Systems

G. Pulley Systems

1. 1 to 1 Pulley System

This is the simplest of all mechanical advantage systems and consists of only a change of direction of pull through a range of devices.

A 1 to 1 system:

- Requires the same amount of force to lift a load as it would have without the pulley system.
- Requires the worker to pull 1 m on the slack end in order to lift the load 1 m.
- Is only practical for loads up to 40 kg.

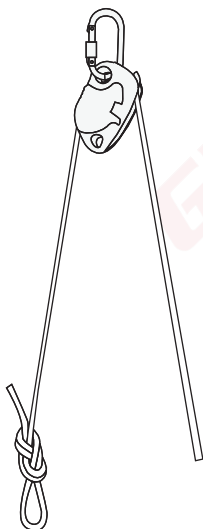
A 1 to 1 Pulley System with a Pulley

- Unsafe. Allows worker to lift the load, but without any control; should the worker let go of the slack rope the load will drop.
- Requires little force due to the pulley that reduces friction.



A 1 to 1 Pulley System with a Progress Capture Device

- Safe. Allows worker to lift the load, but not lower.
- Requires little force as the progress capture device contains a pulley that reduces friction.
- Safe working load is reduced due to the aggressive nature of the progress capture cam.



A 1 to 1 Pulley System with a Descending Device

- Safe. Allows worker to lift and lower the load.
- Requires a lot of force to the added friction of the descender.

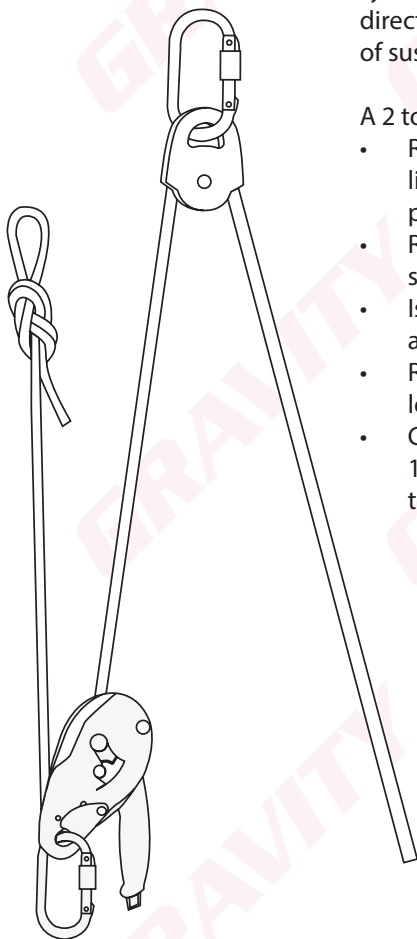


2. 2 to 1 Pulley System

A 2 to 1 pulley system is a little more complicated mechanical advantage system and consists of a change of direction of pulley with the added benefit of suspending the load through 2 ropes.

A 2 to 1 system:

- Requires half the amount of force to lift a load as it would have without the pulley system.
- Requires the worker to pull 2 m on the slack end in order to lift the load 1 m.
- Is practical for loads between 20 kg and 50 kg.
- Requires a backup system for heavier loads.
- Cannot not be converted back into a 1 to 1 system should the worker want to use less rope when lowering.

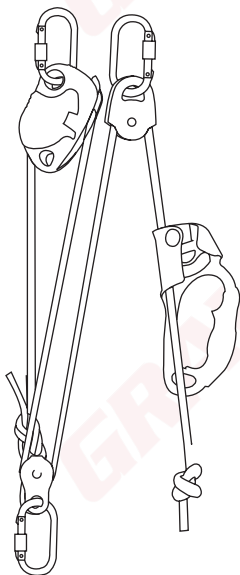


3.3 to 1 Pulley System

A 3 to 1 pulley system is even more complicated, can be configured in one of two ways, and is arguably the most used mechanical advantage system.

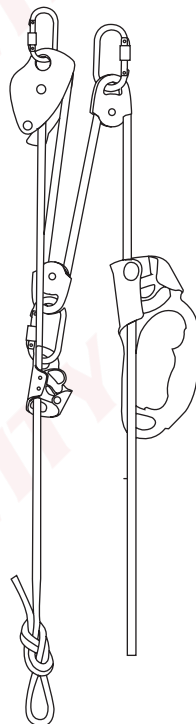
A 3 to 1 system:

- Requires a third of the amount of force to lift a load as it would have without the pulley system.
- Requires the worker to pull 3 m on the slack end in order to lift the load 1 m.
- Is practical for loads between 50 kg and 100 kg.
- Requires a backup system for all loads.



A 3 to 1 Pulley System Rigged Directly to Load

- Allows for a fast lift due to the fact that the system does not need to be reset.
- Uses 3 times the amount of rope a straight lift would have.
- Requires the worker to be next to the load in order for him/her to change it to a 1 to 1 system to lower the load.



A 3 to 1 Pulley System Rigged to a Ascender and a Pulley

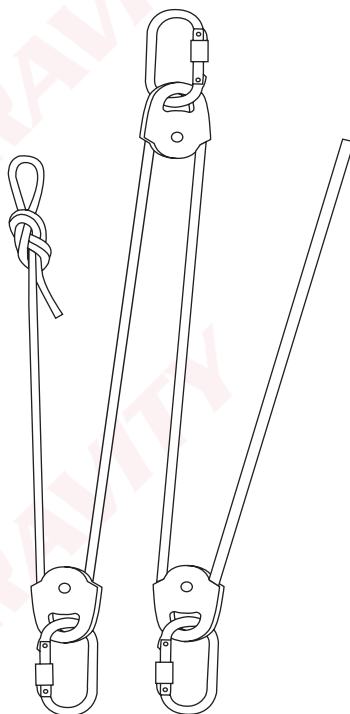
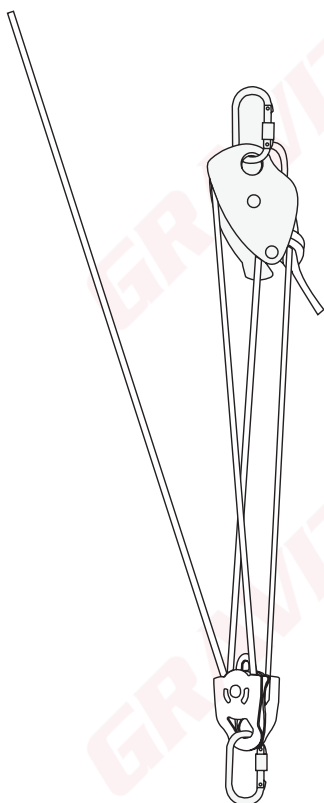
- Slows down a lift due to the fact that the system needs to be reset.
- Uses a smaller amount of rope than what a system rigged directly to the load does.
- The worker can change it to a 1 to 1 system to lower the load easily, without being next to the load.

4. 4 to 1 Pulley System

A 4 to 1 pulley system is seldom used due to the large amount of rope needed in order for it to work effectively, hence, it is only used to lift heavy loads a small distance, e.g. a rescue with a heavy casualty in ascending mode.

A 4 to 1 Pulley System Rigged to Two Separate Pulleys

- Uses a lot of equipment to give a small advantage over the 3 to 1 system.
- The load can not be lowered through a 1 to 1 in order to save rope due to the nature of the system. Thus, if a worker were to lift and lower a load he/she would need 4 times the distance of the lift in rope length.

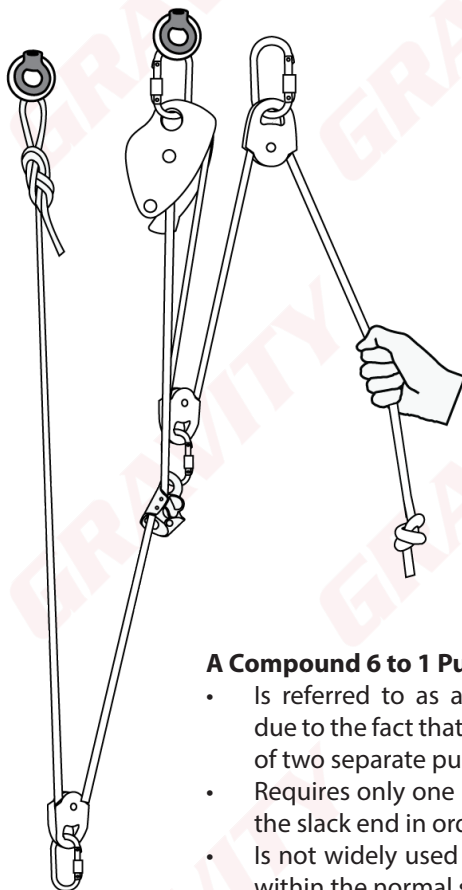


A 4 to 1 Pulley System Rigged to a Single Twin Pulley

- Uses less equipment to give a small advantage over the 3 to 1 system.
- The load can not be lowered through a 1 to 1 in order to save rope due to the nature of the system. Thus, if a worker were to lift and lower a load he/she would need 4 times the distance of the lift in rope length.

4. 6 to 1 Pulley System

A 6 to 1 pulley system is the most complex mechanical advantage system this course covers. It consists of a 3 to 1 system installed on top of a 2 to 1 system. It can also be rigged by continuing to follow the method discussed in the 4 to 1 pulley system rigged to two separate pulleys, but this uses too much equipment to be practical.



A Compound 6 to 1 Pulley System

- Is referred to as a compound system due to the fact that it is constructed out of two separate pulley systems.
- Requires only one sixth of the force on the slack end in order to lift the load.
- Is not widely used due to the fact that within the normal scope of rope access work, a 3 to 1 system will suffice.

5. Calculating the Ratio of a Pulley System

The more complex pulley systems become, the more difficult it will be to see what the ratio of the system is, and therefore, what the mechanical advantage is.

There are two ways one can do this:

5.1. The Simple Pulley Method

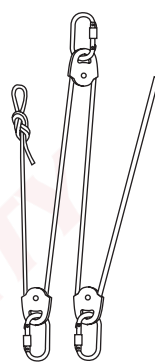
If there are no rope grab devices in the system, one can easily calculate the system by simply counting the amount of ropes running to the load.



1 rope to load
1:1 Pulley system



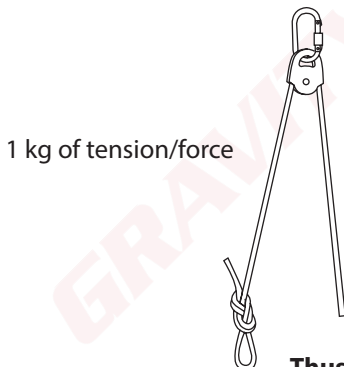
2 ropes to load
2:1 Pulley system



4 ropes to load
4:1 Pulley system

5.2. Follow the Tension

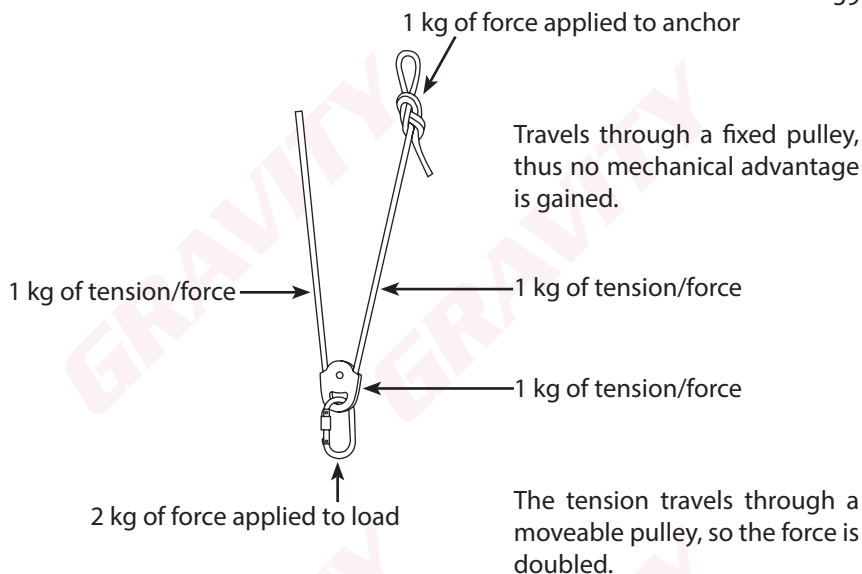
One can also determine the pulley system by following the tension in the ropes. In this case, it is important to remember that one only gains mechanical advantage through a moveable pulley, i.e. a pulley that is not attached to a fixed anchor point.



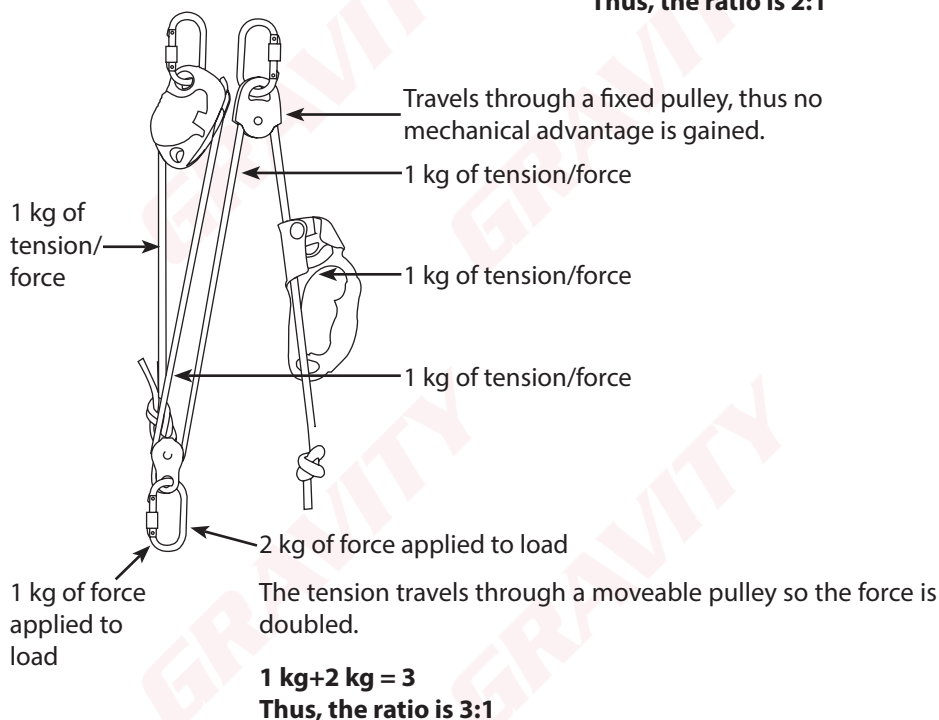
Travels through a fixed pulley, thus
no mechanical advantage is gained.

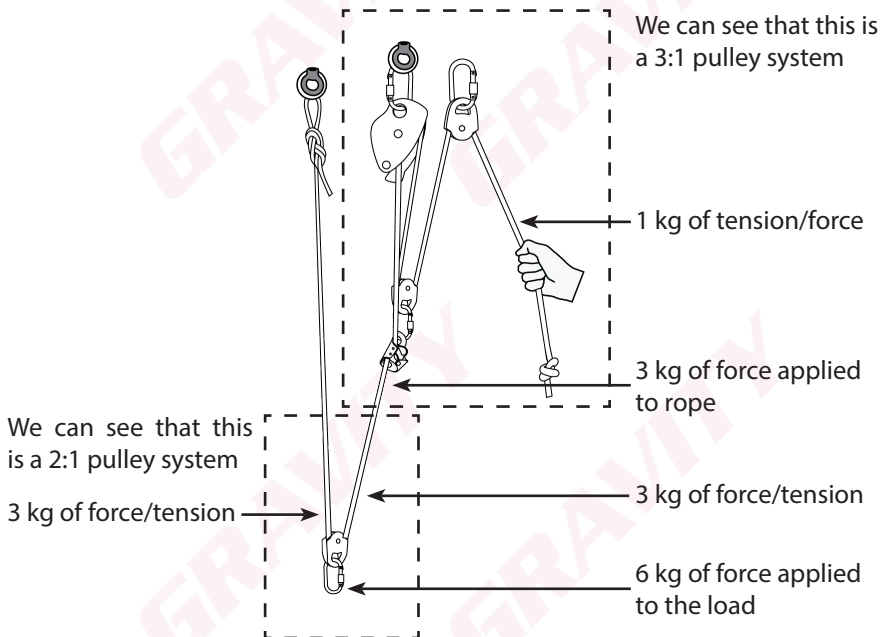
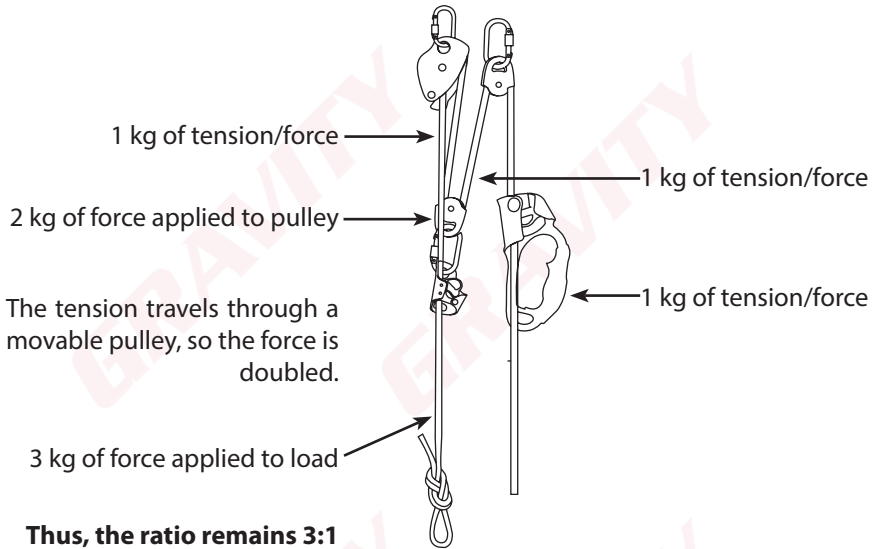
1 kg of tension/force

Thus, the ratio remains 1:1



Thus, the ratio is 2:1





Thus, the ratio is 6:1

H. Emergency Procedures

1. Writing a Rescue Plan

REMEMBER: Prevention is better than the cure. It is the duty of the Supervisor of a rope access site to ensure that incidents do not occur. Being able to perform a rescue out of any possible scenario is important, but ensuring that a rescue does not need to be done has much more value.

When writing a rescue plan, one must ensure that:

- The rescue can be performed at any time, anywhere on site.
- The rescue must be able to be completed within as little time as possible as to minimize the effect of suspension trauma.
- The rescuers are trained to perform required rescues.
- The rescue equipment used is safe for use and fit for purpose.
- The rescue plan must be written in such a way as to not add risks to the site i.e. the rescuers must be protected from fall risks while performing the rescue.

Consider the following when deciding on the most appropriate rescue:

- From what structure/work platform can a worker fall?
- In what position (relative to the point of safety) will the worker be after a fall has occurred?
- What will the rescuer need to reach the fallen worker, e.g. telescopic stick to connect rescue system?
- How many rescuers will be needed?

This will lead you to the following conclusions:

- The type of anchor points that are required to anchor the rescue system, e.g. planned or self-identified.
- The type of rescues that will be required, e.g. lifting/lowering/sideways rescues.
- The additional training (if any) the rescuer must undergo, e.g. approaching the fallen worker via boat.
- The way in which the duties can be divided.

2. First Aid

During an emergency situation on site, remember the principles of first aid, namely:

H - Hazards - analyze the site to ensure that it is safe to enter for you as rescuer.

H - Hello - determine whether the casualty is conscious or not.

H - Help - contact the emergency services.

C - Circulation - determine whether the casualty has a pulse.

A - Airway - determine whether the casualty's airway is obstructed or not.

B - Breathing - determine whether the casualty is breathing.

The "HHH"-phase should always be the first port of call for any rescuer before attempting a rescue. The information gathered in this phase will determine much of how the rescuer should proceed.

The "CAB"-phase of the procedure will vary from incident to incident and therefore, it is of utmost importance that all rescue technicians, regardless of their competence level, should be well versed in first aid procedures.

A rescuer will be called on to make difficult judgement calls once engaged in the rescue, for example: if a rescuer arrives at a casualty who is bleeding, it will be the choice of the rescuer whether to apply first aid before or after lowering or lifting the casualty to safety.

As always, should a casualty be in need of CPR, this should be applied without hesitation, if at all possible.

Should there be none of these complications, the rescuer can begin looking for and treating secondary injuries. Most likely these are fractures, which should be stabilised (if possible) before attempting to move the casualty as this will save the casualty a considerable amount of pain and reduces the chance of possible further harm.

Should the casualty have serious fractures, or if the rescuer suspects spinal injuries, it is recommended that the rescue be done with the casualty secured in a stretcher.

Shock will most likely also be present, as suffering a fall is a traumatic experience. Remember to treat for shock during and after the rescue, monitoring the condition of the patient until the emergency services arrive and take control.

3. Principles of Rescue

Although each rescue scenario will be different, there are a few principles that must be adhered to during all rescues in order for both the rescuer and casualty to be safe.

1. Rig for rescue as far as reasonably practical.
2. The rescuer must always be connected to two points.
3. The casualty must always be connected to two points.
4. The casualty may never be suspended on the rescuer's harness whilst the rescuer is on ascending equipment.
5. The rescuer may never be suspended on the casualty's harness whilst the casualty is on ascending equipment.
6. Never shock load any part of the rescue system.
7. Never allow the system to be set up in such a way that a shock load would damage the system.
8. No uncontrolled swings or descents of the rescuer or casualty are allowed.
9. All load bearing karabiners must be closed and locked, as well as loaded on the spine.
10. Never use equipment outside of manufacturer's recommendations.
11. Never attempt a rescue beyond the scope of your competency.
12. Always ensure that the correct equipment is on site in order to perform an efficient rescue with as little wasted time as possible.
13. A rescue plan should be in place for each and every scenario in which a worker may find him/herself on site. This plan should be communicated to all team members, regardless of competency, in order to allow for as many helpers as possible should the need arise. All these preparations and preparations needed for a rescue should be done before work starts.

As a rescuer, and a rope access supervisor, it is your duty to ensure that you and your team remain fit and well versed in all techniques and procedures needed to establish and maintain a safe work environment.

4. Fall Arrest Rescue

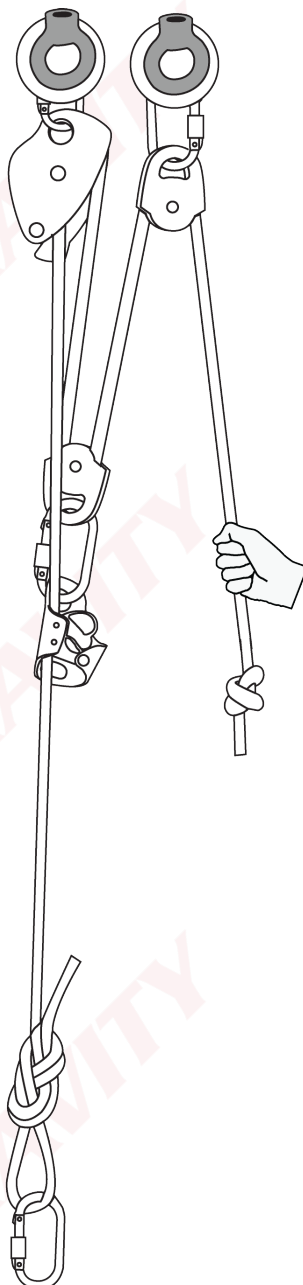
3:1 Rescue

Rescue Procedure

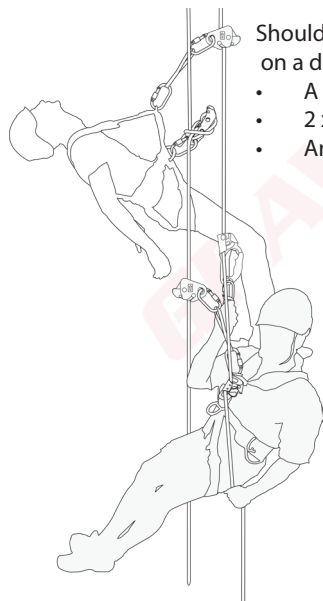
1. Call the emergency medical services.
2. Climb up to the casualty.
3. Connect the rope to casualty's top D-link.
4. Connect a suspension trauma sling.
5. Climb above the casualty.
6. Anchor the descending device to structure.
7. Connect the Croll and pulley to rope.
8. Connect the rescue pulley to structure.
9. Ensure all slack is removed between the casualty and the descending device.
10. Lift the casualty.
11. Remove the casualty's lanyard.
12. Remove the 3:1 system.
13. Apply extra friction.
14. Lower the casualty.

When performing a rescue, remember that communication is of utmost importance and **MUST BE MAINTAINED AT ALL TIMES.**

To save time when attempting a rescue, as well as to reduce the risk of errors occurring and/or dropping items when at height, it is always a better idea to rig the rescue system when on the ground before climbing up to the casualty.



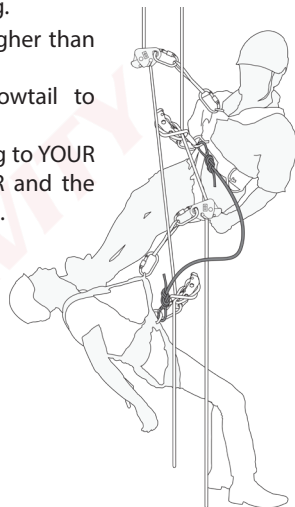
5. Rescue from a Descending Device



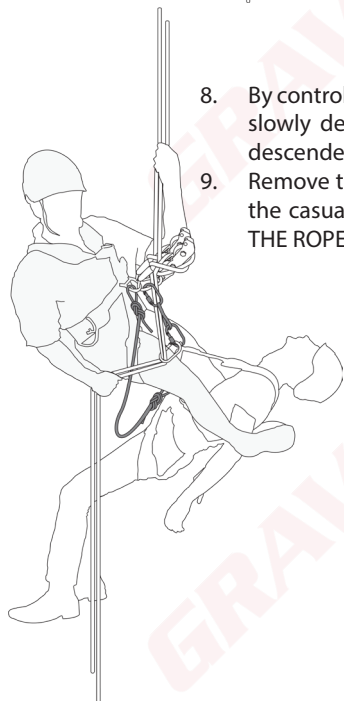
Should a worker be unable for any reason to reach the ground whilst on a descending device, a rescue must be performed. You will need:

- A rescue sling
- 2 x Karabiners
- An extra locker (to be connected to a cowtail):

1. Connect your backup device on the casualty's main line and vice versa.
2. Ascend until you are the same level as the casualty.
3. Connect your cowtail locker above the casualty's stop.
4. Connect your locker sling to the locker above the casualty's stop and remove your cowtail.
5. Connect the casualty's cowtail to his/her locker and remove their locker sling.
6. Ascend until you are higher than the casualty.
7. Connect your extra cowtail to their BOTTOM D-LINK.
7. Connect the rescue sling to YOUR DESCENDER KARABINER and the CASUALTY'S TOP D-LINK.



8. By controlling the casualty's descender, slowly descend the casualty until the descender becomes slack.
9. Remove the casualty's descender from the casualty's harness BUT NOT FROM THE ROPE.



10. Connect the casualty's descender to your harness; ENSURE THAT THE GATES OF BOTH KARABINERS ARE FACING DOWNWARD AND TOWARDS YOU, and remove all the slack and lock it.
11. Remove your back-device.
12. Use the karabiner connecting the casualty's locker sling to their harness and connect this karabiner to the slack ends of BOTH ropes.
14. Whilst keeping tension on the slack ends of the rope running through the extra karabiner, descend down in a controlled fashion.
15. Open both karabiners and remove your BOTTOM D-LINK.

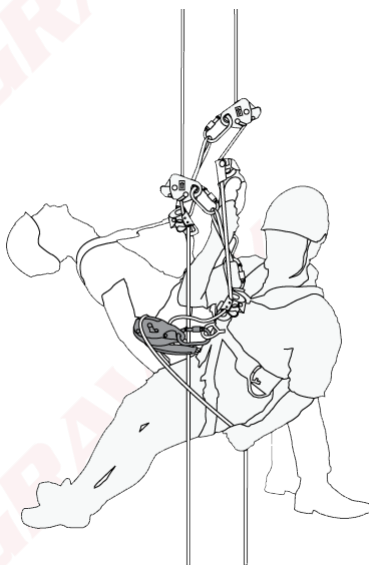
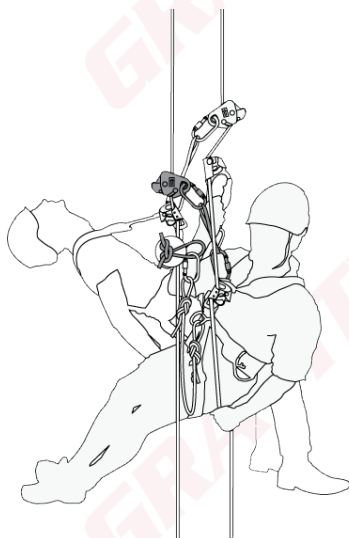
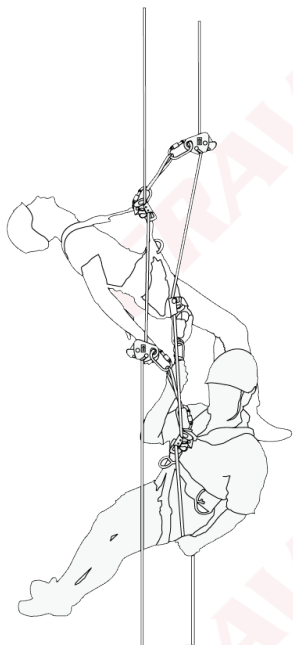
6. Rescue from Ascending Equipment (pick-off rescue)

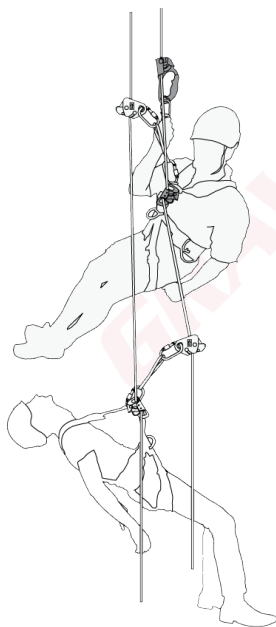
Should a worker be unable, for any reason, to reach the ground whilst on ascending equipment, suspended on a vertical lifeline or stuck on a work positioning lanyard, a rescue must be performed. You will need:

- A rescue sling
- 3 x Karabiners (2 of these karabiners are attached to the rescue sling)
- A cheat-string (this can be substituted with a cowtail if need be).

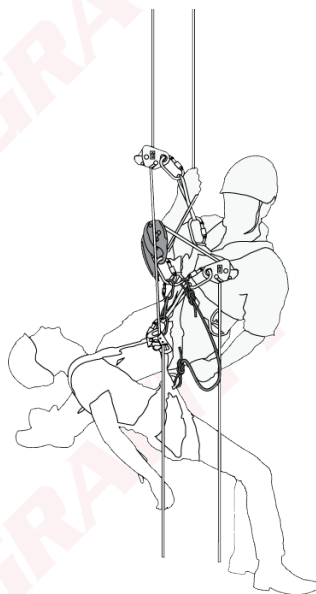
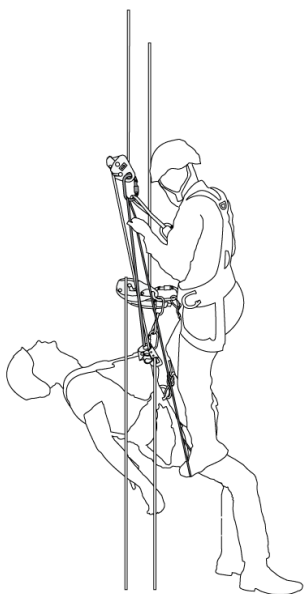
Remember the Principles of Rescue

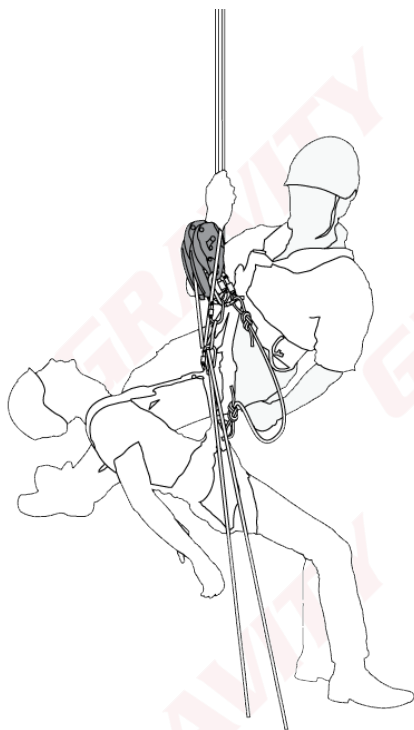
1. Connect your backup device on the casualty's main line and vice versa.
2. Ascend until you are at the same level as the casualty.
3. Move past the casualty as though he/she were a knot in the rope.
4. Once above the casualty, perform a change over.





5. Once above the casualty and on descending, you can now connect the extra cowtail (either the casualty's or your own, however, it is preferable to use your own in case of emergency) to his/her bottom D-link.
6. Connect a rescue sling between your descender karabiner and the casualty's top D-link.
7. Using the cheat-string/cowtail, connect onto the casualty's chest ascender (if this is not possible, connect to the top D-link).
8. Thread the other end of the cheat-string/cowtail through the karabiner of your backup device.
9. Using either the handled ascender, or by making a loop in the cheat-string/cowtail, apply your entire body weight to the cheat-string/cowtail.
10. This should allow you to lift the casualty quite easily with one hand in order to disengage the chest ascender.



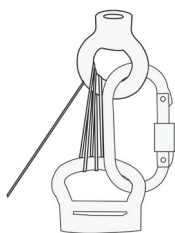


11. Install a second descender (either your's or the casualty's) on your backup line.
12. Remove your backup device.
13. Apply extra friction.
14. Descend down in a controlled fashion.

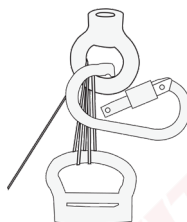
Performing a Pick-Off from a Short-Link such as a Karabiner or Locker Sling



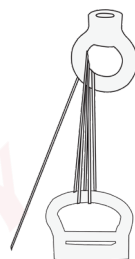
1. A casualty is suspended on a short-link between an anchor point and his/her harness.



2. Wrap a cheat-string multiple times through the harness and anchor point.



3. Apply your body weight to lift casualty and to enable the removal of the short-link.



4. Lower the casualty down slowly onto which ever device is being used for rescue.

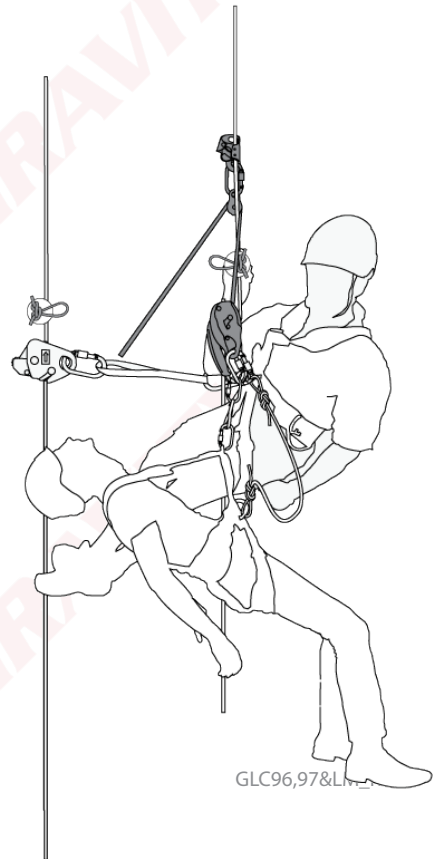
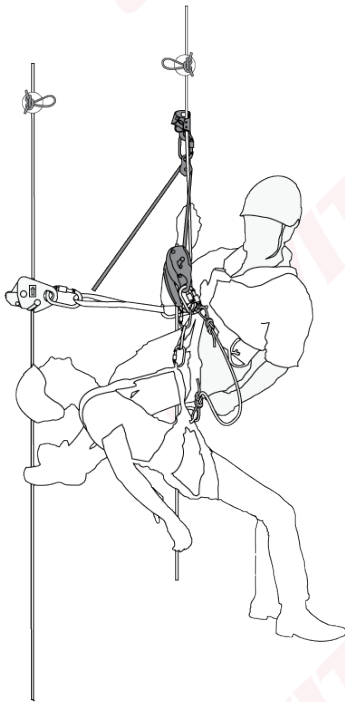
7. Ascending with a casualty (past knots)

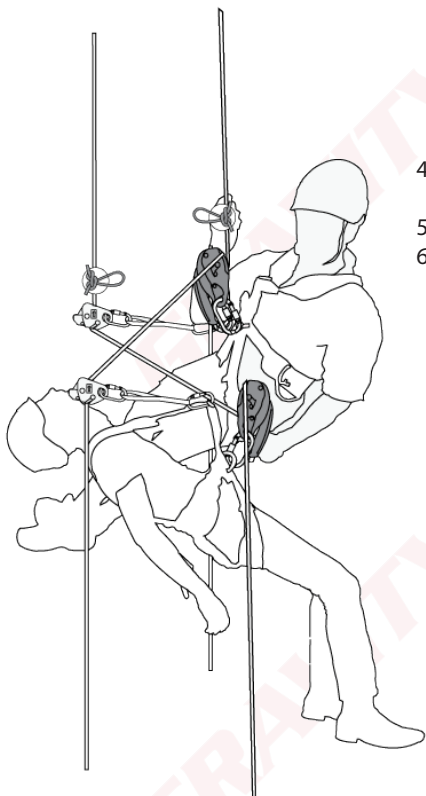
Should a worker be suspended or incapacitated below knots in the rope with the point of safety above, a rescue must be performed. You will need:

- A rescue sling
- 4 x Karabiners (2 of these karabiners are attached to the rescue sling)
- A rope grab
- A pulley

Remember the Principles of Rescue

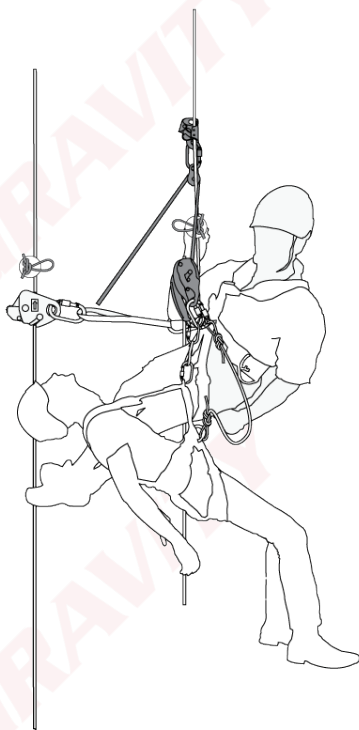
1. Connect the casualty to you with a rescue sling and an extra cowstail.
2. Connect the pulley system above your descender.
3. Ascend until the knot is as close as possible to the descender.





4. Connect the casualty to his/her own descender and back-up.
5. Descend until the rescue sling becomes slack.
6. Remove the rescue sling and the extra cowstail.

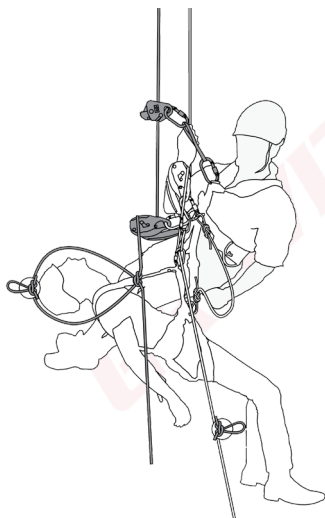
7. Jump the knot as per normal level 1 procedure.
8. Once above the knot, reconnect the rescue sling and extra cowstail.
9. Remove the casualty's descender and back-up device.



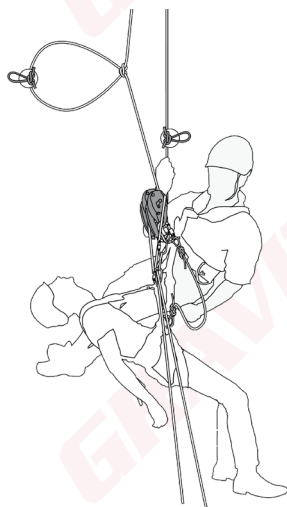
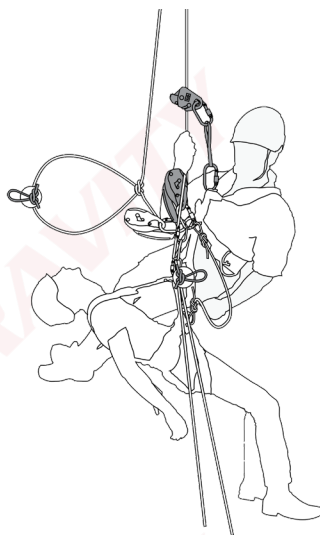
8. Rescue Pass Knots

Remember the Principles of Rescue

1. The casualty is attached to the rescuer either by performing rescue off descending or ascending equipment.
2. Select one of the knots in the ropes and either adjust the knot (if it is an Alpine Butterfly), or make a bigger Alpine Butterfly ensuring that the original knot is inside the loop of the Alpine Butterfly.

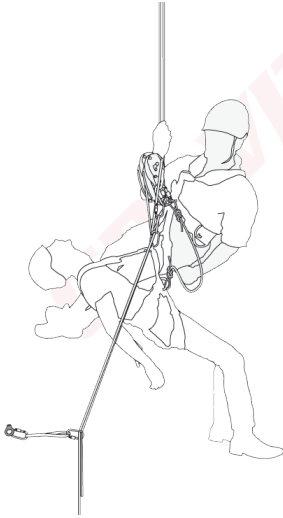


3. Descend until one descender is approximately 30 cm above the knot.
4. Install a backup device above the descender.
5. Descend with the descender under the backup device until it becomes slack.

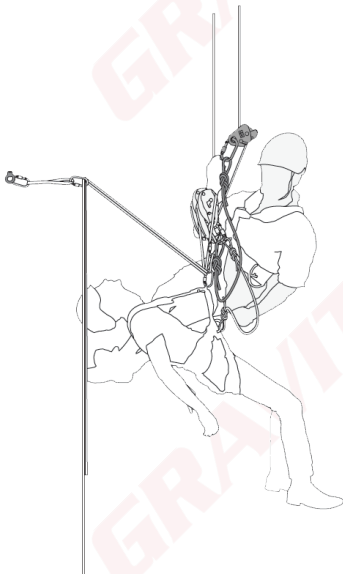


6. Move the descender underneath the knot.
7. When the second knot is reached, repeat the process and move the second descender underneath the knot.

9. Deviation Rescue

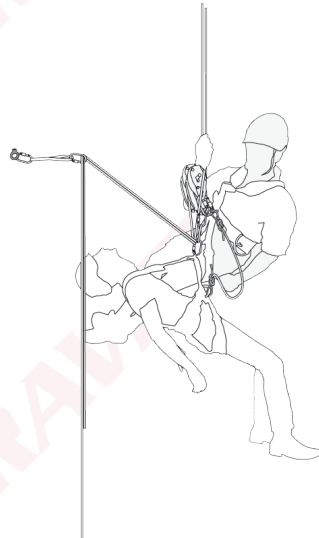


2. Descend until the rescuer's eyes are level with the deviation anchor.

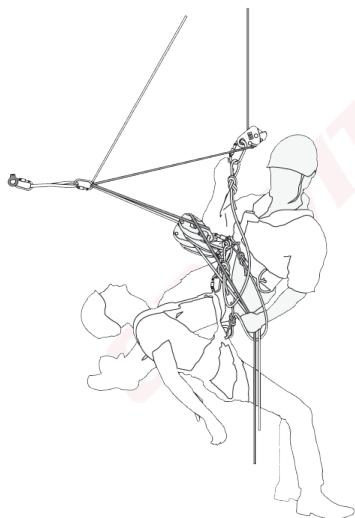


Remember the Principles of Rescue

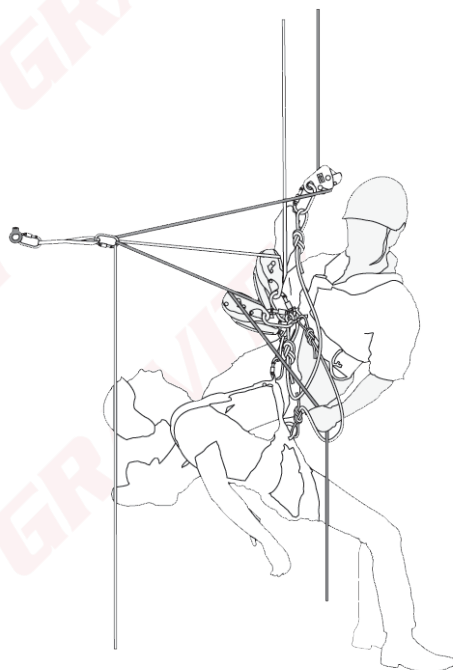
1. Attach the casualty to the rescuer by performing a pick-off rescue, either off of ascending or descending equipment.



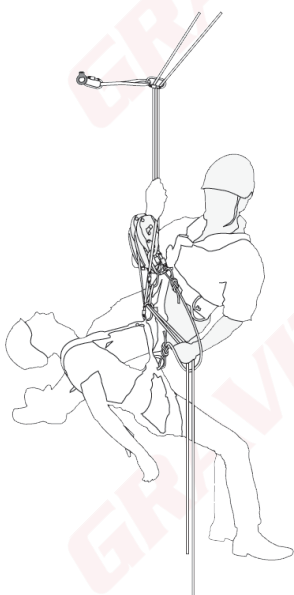
3. Install the backup device on a cowtail above one of the descenders.
4. Descend one of the descenders until it becomes slack.
5. Remove the slack descender and install it on the same rope underneath the deviation anchor point.
6. Pull the rope through the descender until it is as tight as possible.



7. Move the backup device to the other rope.



8. Repeat steps 4-6.
9. Remove the backup device.



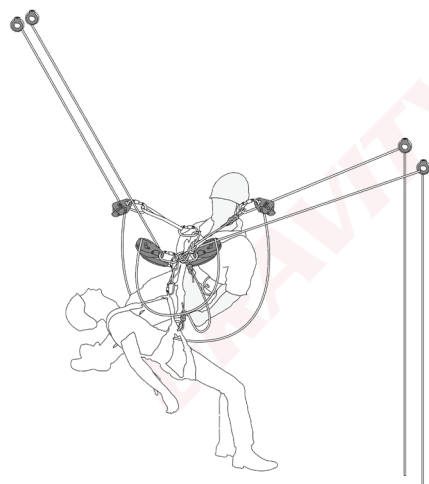
9. Apply extra friction and descend down in a controlled fashion.

Please Note: the steps for a rescue passing a DOUBLE DEVIATION the steps remain the same.

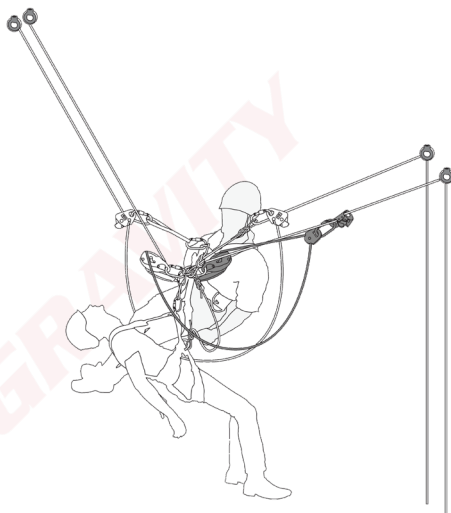
10. Re-belay/re-anchor Rescue (with re-belay less than 1.5m)

Remember the Principles of Rescue

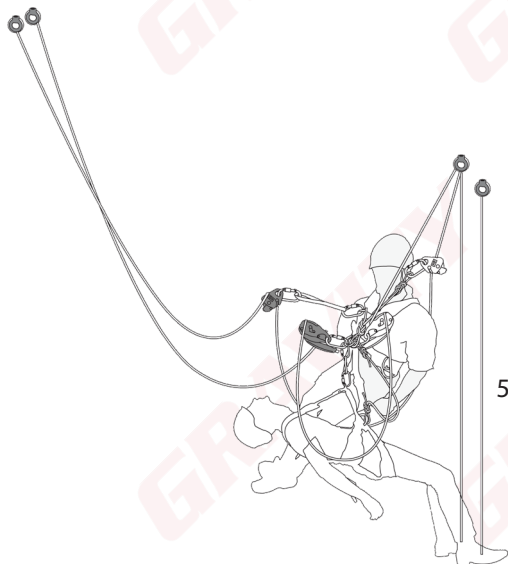
1. Attach the casualty to the rescuer by performing a pick-off rescue, either off of ascending or descending equipment.
2. Attach a second descender and a second backup device on the loop towards the straight drop.

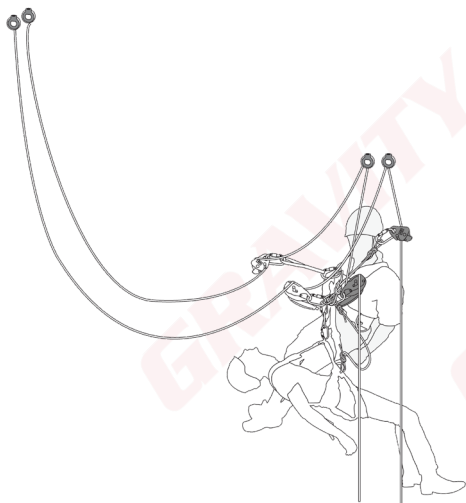


3. Attach a 2 to 1 pulley system using the rope of the descender rope closest to the straight drop.
4. Ascend and descend until all weight is on the devices closest to the straight drop.



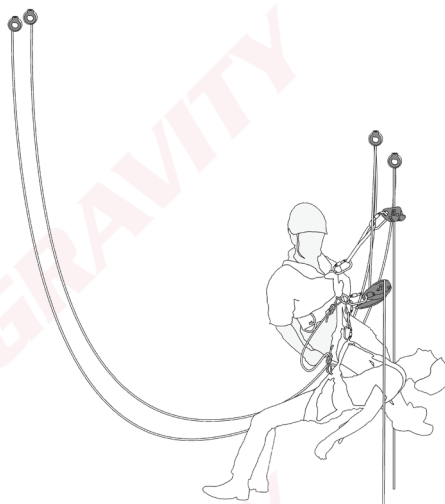
5. The trailing devices will become slack; remove them from the loop.



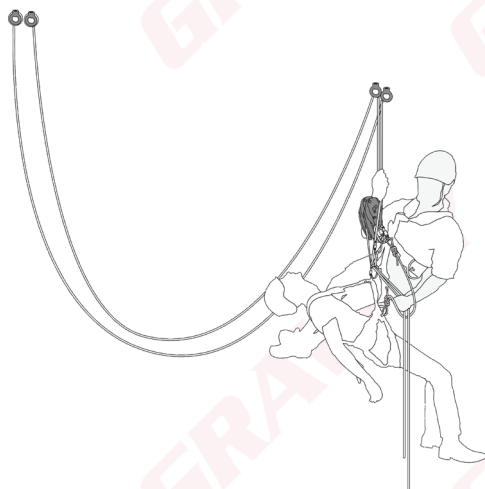


6. Move the removed devices from the loop onto the straight drop.

7. Transfer the weight from the loop onto the straight drop.
8. Remove all equipment from the loop.



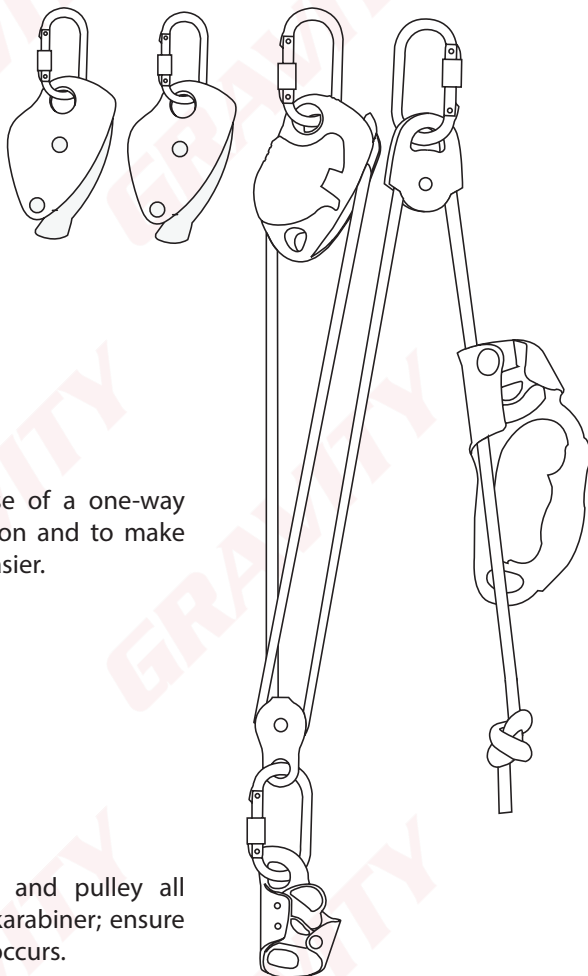
9. Install a second descender on the straight drop and remove the backup device.
10. Apply extra friction and descend in a controlled fashion.



11. Plug-in Rescue

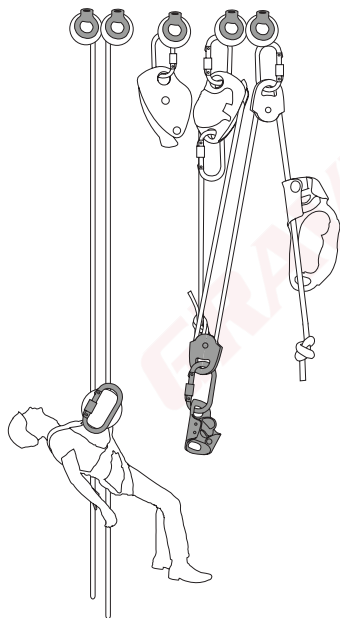
Below is an example of how to assemble a plug-in system. The karabiners can either be connected to planned anchor points (if available), slings or a rigging plate.

The plug-in system consists of a 3 to 1 system rigged in such a way that the connection point is a rope grab device rather than the traditional knot and karabiner.



The system makes use of a one-way pulley to reduce friction and to make the lifting operation easier.

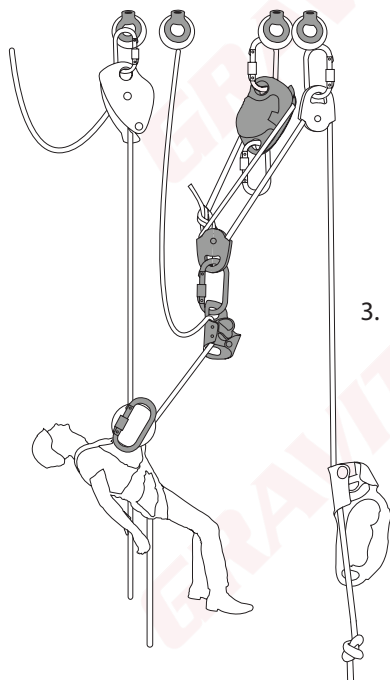
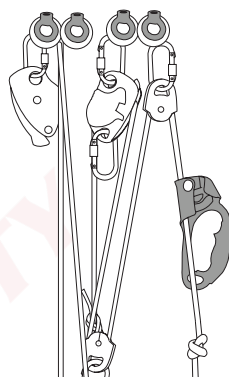
The knot, rope grab and pulley all connect to the same karabiner; ensure that no cross-loading occurs.



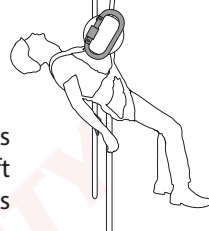
A plug-in rescue is required should the ropes that the casualty is suspended on be tight and there is no option of reaching the casualty.

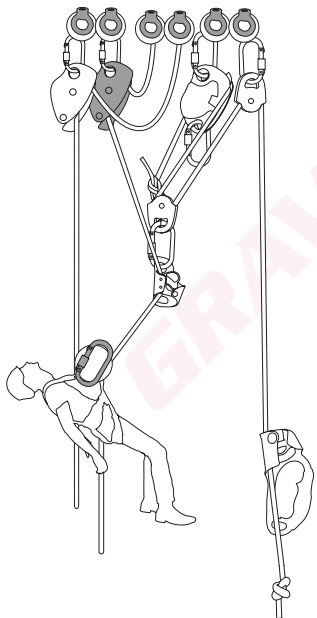
Remember the Principles of Rescue

1. A 3 to 1 system is rigged. Notice that the pulley system ends with a rope grab device (croll) and not a knot as usual.
2. Install the pulley system as far down on the rope as possible.



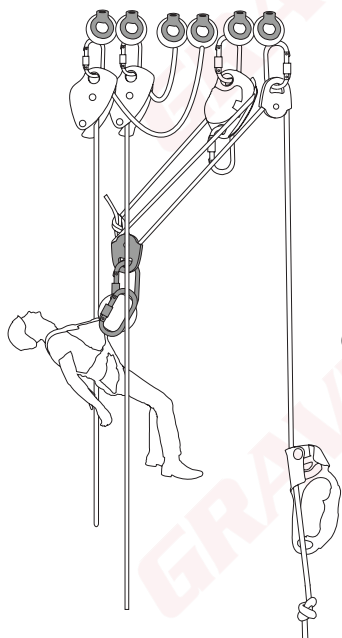
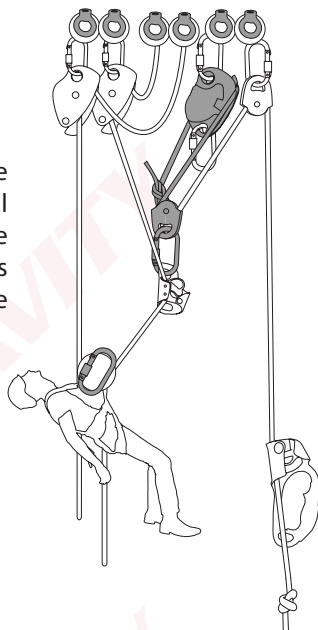
3. Engage the progress capture device and lift the casualty as far as possible.



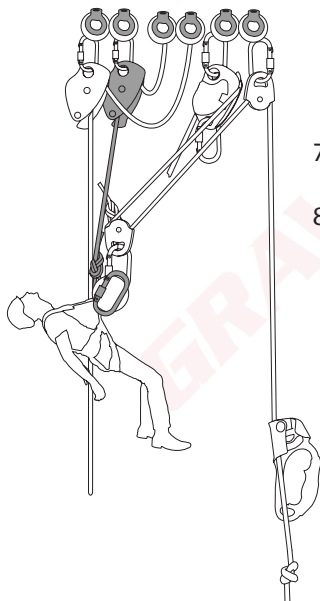


4. Use the slack in the ropes that was created in step 3 to install two descending devices on the casualty's ropes.

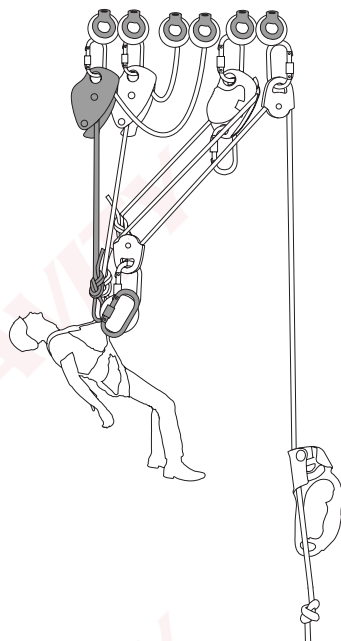
5. Continue to lift the casualty. Remove all slack from the descending devices as you lift. Reset the system as needed.



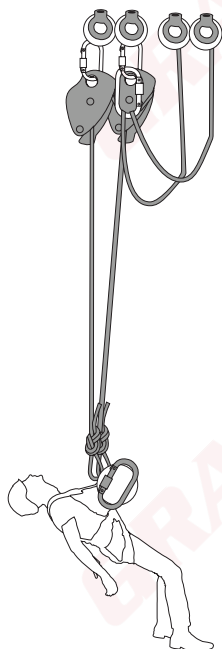
6. When the casualty can be reached, install the plug-in system on one of the ropes and lift the casualty until the device that the casualty is suspended from (on that rope) can be removed.



7. Take the bottom of the rope that has been removed and connect it directly to the casualty's harness.
8. Release the tension from the plug-in system until the casualty is suspended on the rope.

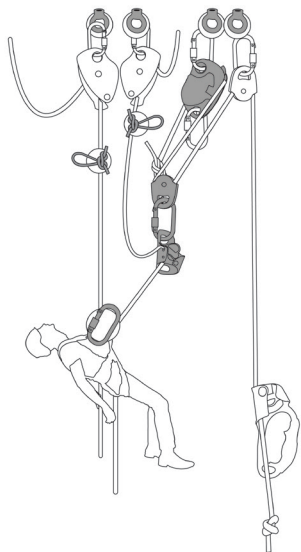


9. Repeat step 6-8 for the other rope.



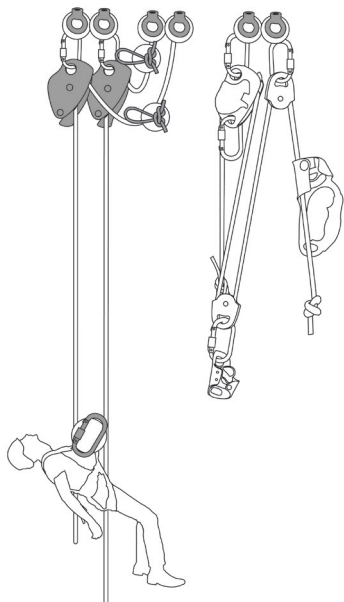
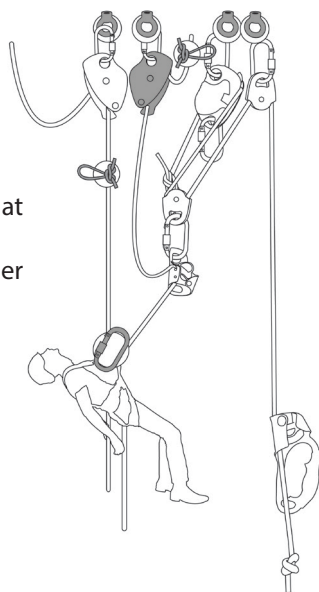
10. Apply extra friction and lower the casualty down in a controlled fashion.

12. Plug-in Rescue Pass the Knots



1. The rescue procedure remains the same as a normal plug-in until a knot reaches a descending device.
2. When the knot needs to be passed, use the plug-in system to lift on the relevant rope in order to create enough slack so that the knot can be passed through the descender.

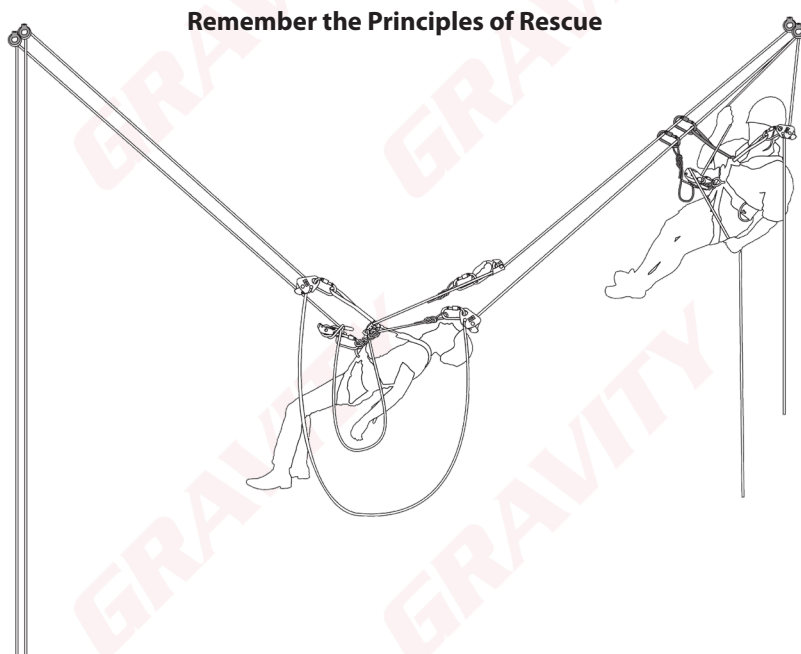
3. Remove the descender and re-install the rope so that the knot is on the other side of the device.
4. This process is repeated when the knot in the other rope is reached.



5. When lowering the casualty past knots, the same procedure is followed. The plug-in system is re-installed, slack is created, and the knot is moved past the descender.

13. Loop Rescue Option 1

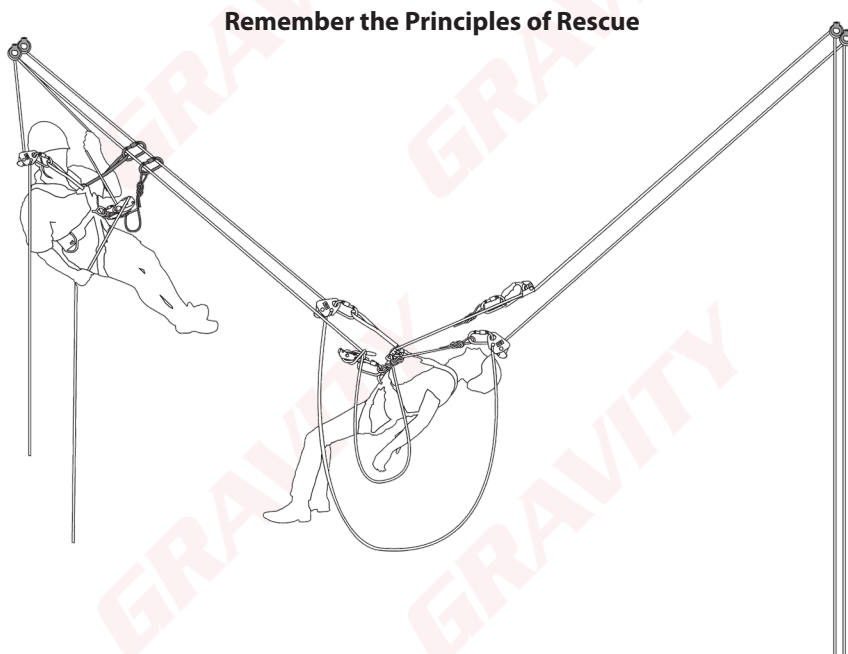
The first option for the loop rescue is the easiest method. This is the preferred method should there be no obstructions or complications preventing use of this rescue.



1. Connect 2 points onto the loop.
2. Descend down the straight drop until the casualty is reached.
3. Connect a rescue sling between the rescuer's and casualty's BOTTOM D-link.
4. Descend the casualty's descender until he/she is hanging straight below the anchor points.
5. Connect an extra cowtail to the casualty.
6. Remove the casualty's descender.
7. Remove the casualty's backup from the descending side.
8. Perform a rescue off ascending equipment.
9. Remove the casualty's backup device from the ascending side.
10. Install the casualty's descender underneath your backup device.
11. Remove your backup device.
12. Apply extra friction and descend in a controlled fashion.

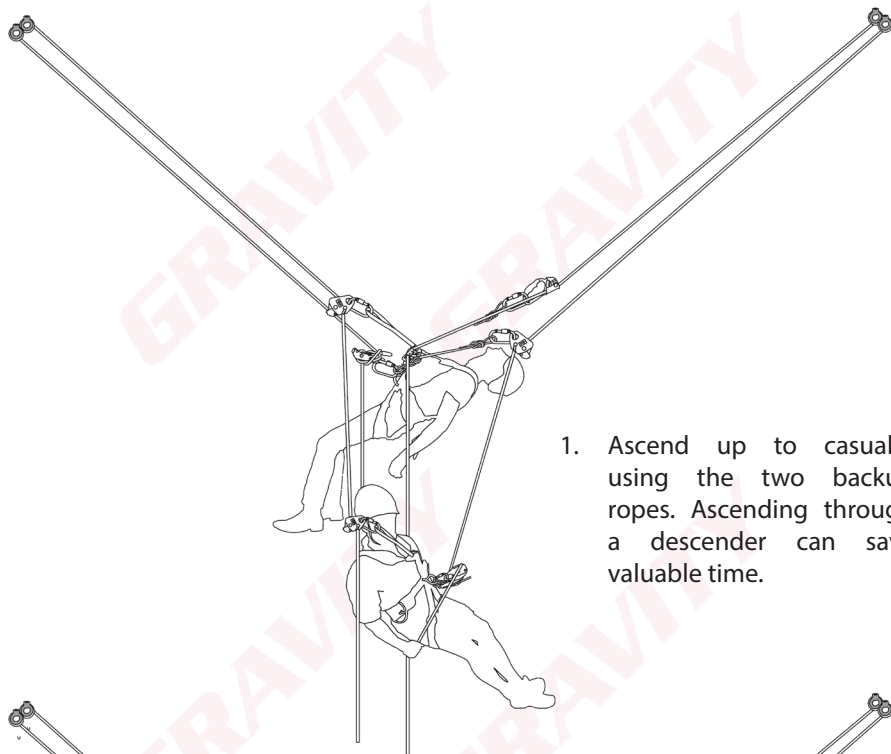
14. Loop Rescue Option 2

Should it be impossible for the rescuer to enter the loop on the casualty's ascending side (perhaps the ropes are too short or there might be an obstruction), the rescuer will have to enter on the casualty's descending side.

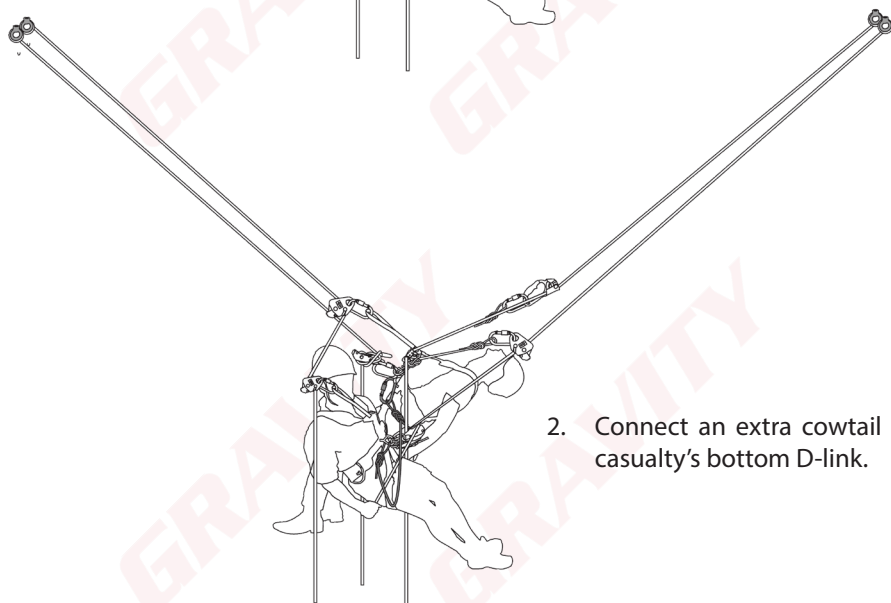


1. Connect 2 points onto the loop.
2. Descend down the straight drop until the casualty is reached.
3. Connect a rescue sling between the rescuer's and casualty's BOTTOM D-link.
4. Connect an extra backup device above the casualty's ascending equipment.
5. Slacken and remove the casualty's descender.
6. Install casualty's descender underneath the casualty's backup device on the ascending side and remove all slack.
7. Remove the casualty's backup from the descending side.
8. Rescue the casualty off ascending equipment.
9. Descend until the rescuer and casualty are hanging straight underneath the anchor points.
10. Remove equipment from loop.
11. Install second descender on straight drop and remove backup device.
12. Apply extra friction and descend in a controlled fashion.

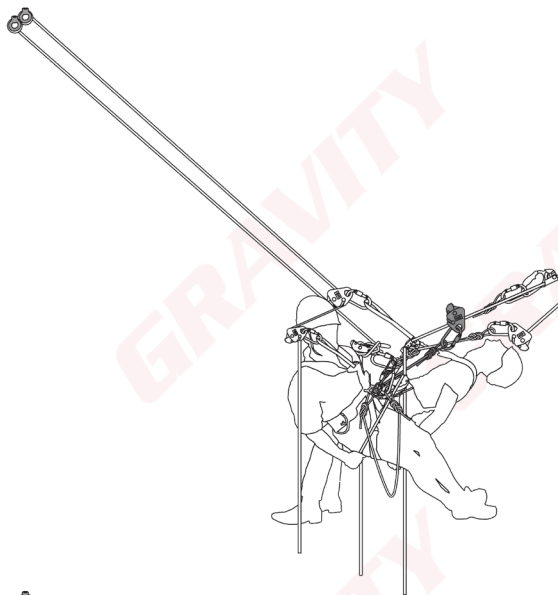
15. Long-rope Transfer Rescue



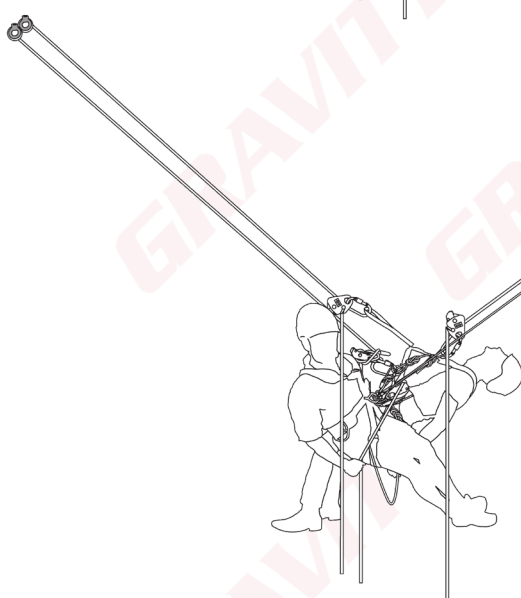
1. Ascend up to casualty using the two backup ropes. Ascending through a descender can save valuable time.



2. Connect an extra cowtail to casualty's bottom D-link.



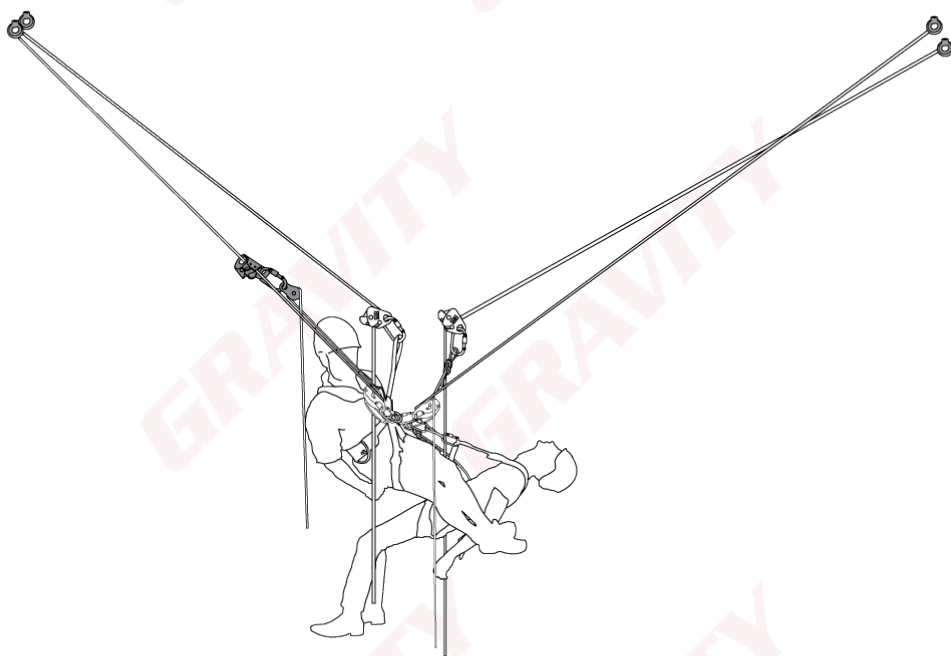
3. Connect an extra backup device above casualty's chest ascender.
4. Connect a rescue sling between the rescuer's descender and casualty's top D-link



5. Remove the casualty's chest ascender. This can be done by either performing a pick-off rescue (which can be difficult), or ascending with the casualty.
6. Remove the casualty's handled ascender.
7. The rescuer now has a choice of which side to descend down from by controlling his/her own, or the casualty's descender.

13. Rope Transfer with Casualty

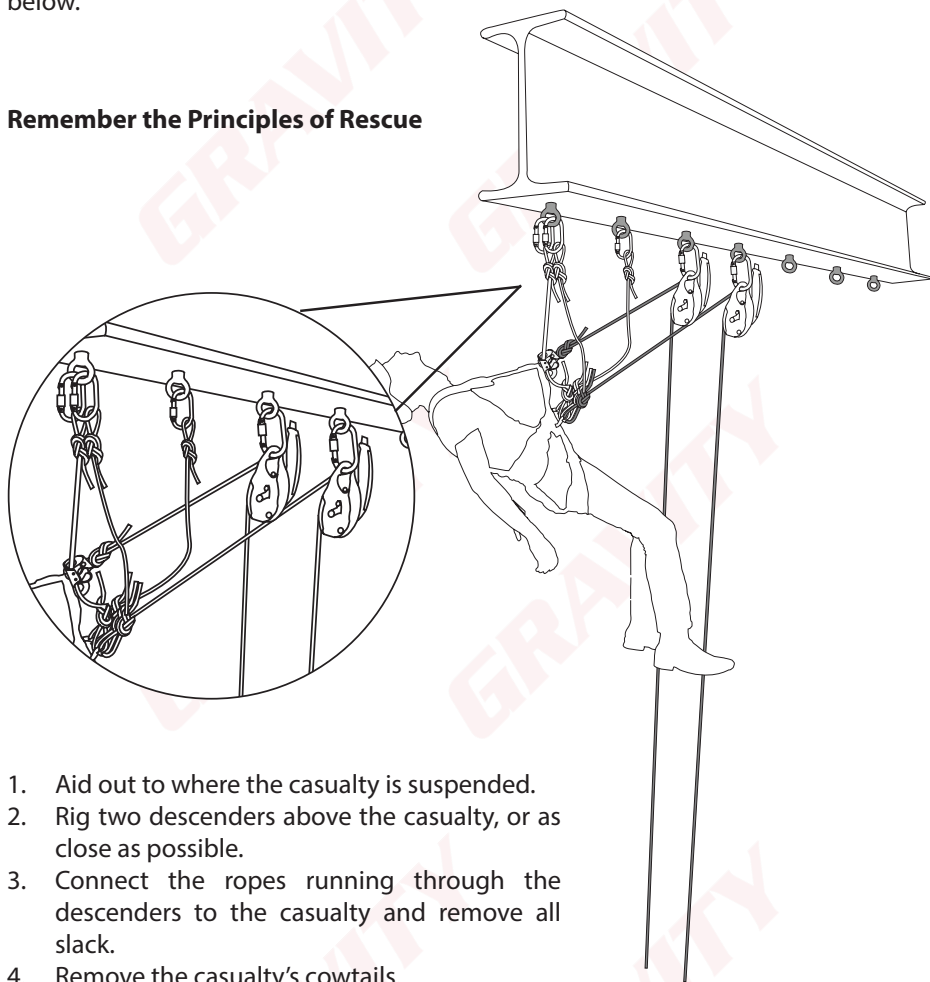
1. Connect the casualty to your harness with a rescue sling and an extra cowstail.
2. Connect a second back-up device and descender on the second set of ropes you wish to transfer to.
3. Connect a pulley system above the second descender.
4. Descend and ascend alternatively until the transfer is complete.



14. Aiding Rescue Option 1

The type of aiding rescue required will depend on whether there is a point of safety below the aid line or not. Below is a rescue in the case where there is a safety point below.

Remember the Principles of Rescue

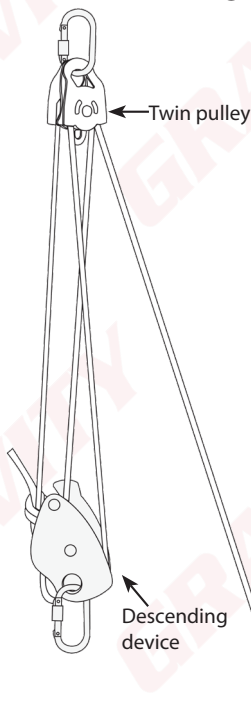


1. Aid out to where the casualty is suspended.
2. Rig two descenders above the casualty, or as close as possible.
3. Connect the ropes running through the descenders to the casualty and remove all slack.
4. Remove the casualty's cowtails.
5. Perform a pick-off rescue with a cheat string to remove the connection point from which the casualty is suspended from.
6. The casualty will now hang on the two descenders.
7. Apply extra friction and lower the casualty down in a controlled fashion.

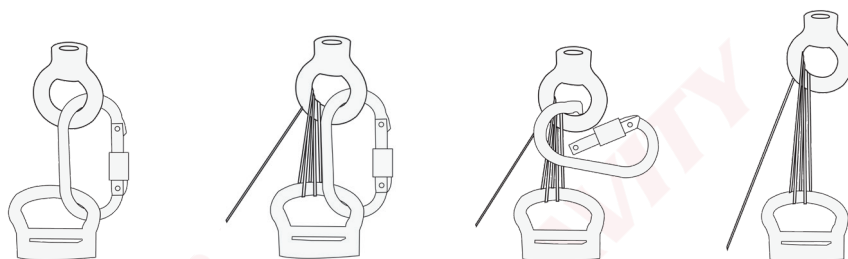
15. Aiding Rescue Option 2

Should it be impossible for the casualty to be lowered straight down, it will be necessary to aid with the casualty to the nearest point of safety. This can be done by using one of two pulley systems.

3 to 1 with a descending device

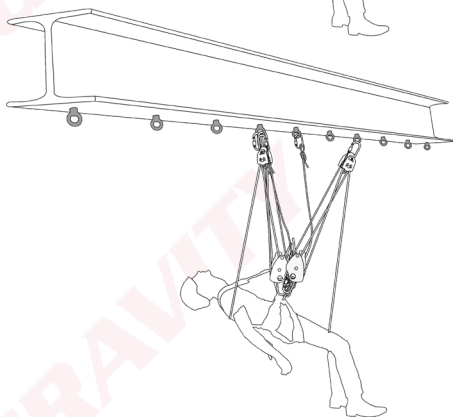
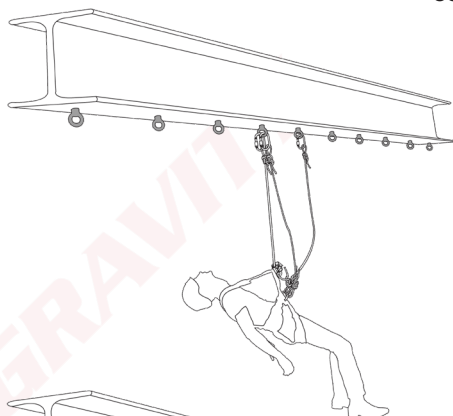
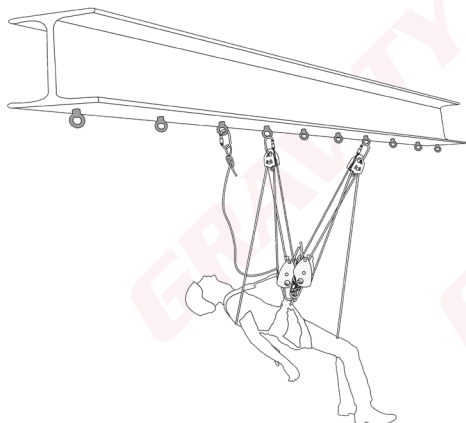


Performing a Pick-Off from a Short-Link such as a Karabiner or Locker Sling

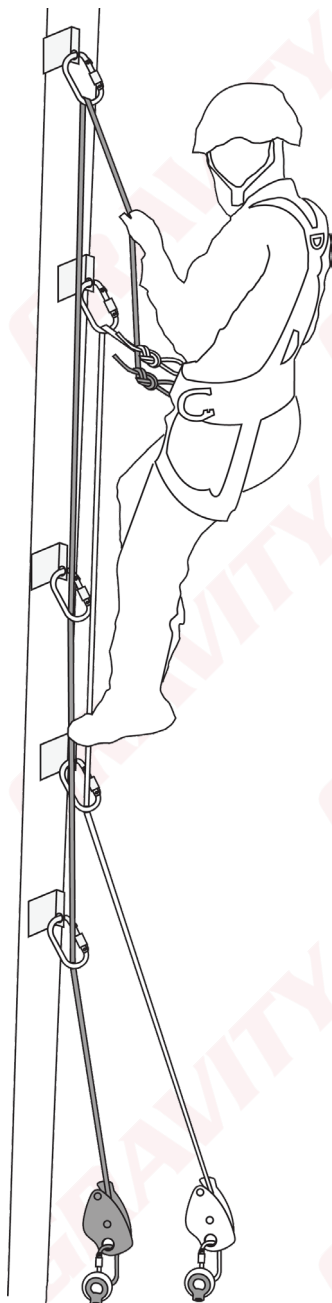


1. A casualty is suspended on a short-link between an anchor point and his/her harness.
2. Wrap a cheat-string multiple times through the harness and anchor point.
3. Apply your body weight to lift casualty and to enable the removal of the short-link.
4. Lower the casualty down slowly onto which ever device is being used for rescue.

1. Connect two pulley systems to the casualty, either 3:1 or 4:1.
2. Lift the casualty in order to remove all but one cowtail.
3. Lift the casualty on the front most pulley system and descend the pulley system at the back.
4. Move the slack pulley system to the front.
5. Move the cowtail forward as a backup.
6. The rescuer aids alongside the casualty in the normal fashion.



Casualty suspended from two 3 to 1 pulley systems.

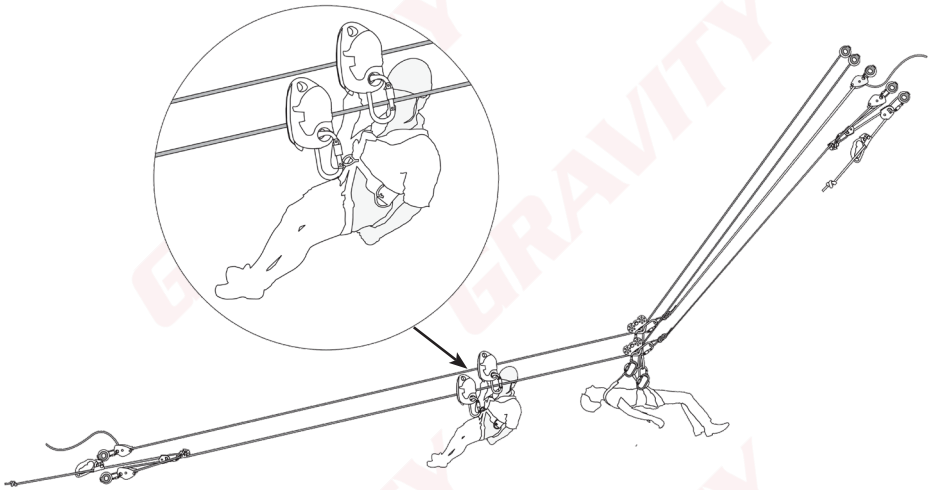


16. Lead Climbing

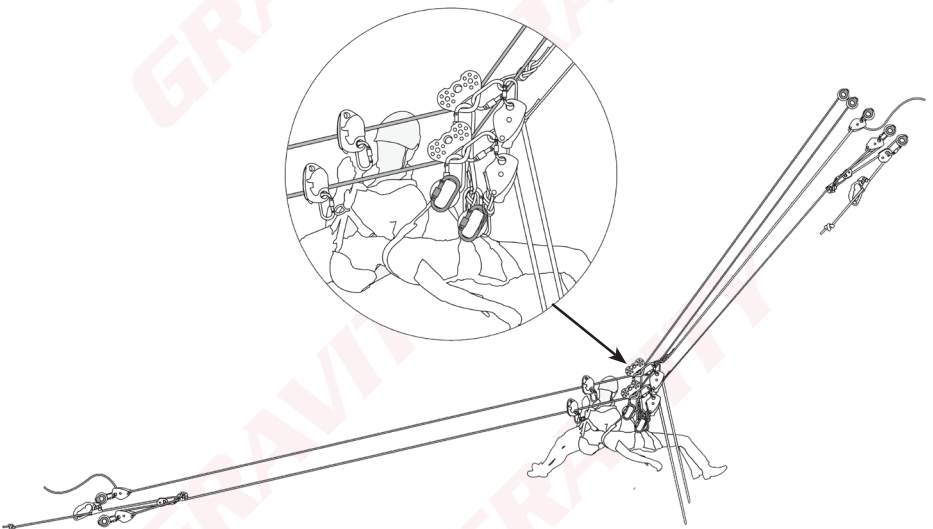
Lead climbing is an alternative to aiding with the added bonus that the rescue can be performed immediately.

It consists of two descenders rigged with dynamic rope, connected to the climber's harness. The climber then alternately clips the two ropes in anchor points.

17. Diagonal Cable Way Rescue

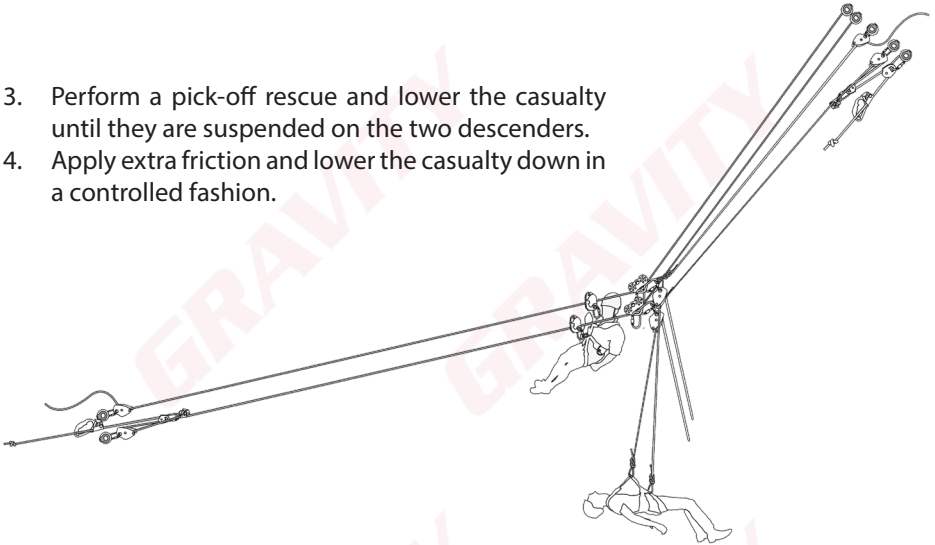


1. The rescuer ascends to the casualty using two progress capture devices. (Pro-traxions in the example given.)



2. Connect 2 descenders to the tagline's Figure 8.

3. Perform a pick-off rescue and lower the casualty until they are suspended on the two descenders.
4. Apply extra friction and lower the casualty down in a controlled fashion.



I. Knots

1. Knot Strengths

All knots weaken ropes. As a rope access supervisor, you should know how much each knot weakens a rope in order to better know the limitations of your system.

It is also best practice to not leave knots in a rope for longer than six months as this can severely weaken a rope. The best solution would be to re-tie knots daily.



Figure 8 on a Bight = 80% of rope strength

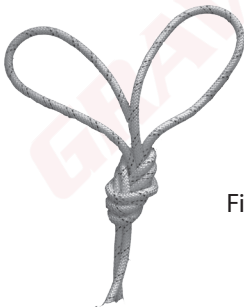


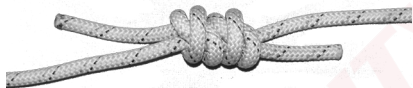
Figure 9 on a Bight = 80% of rope strength



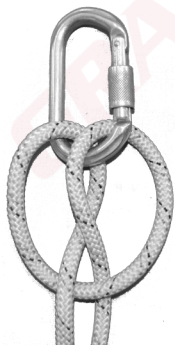
Figure 9 on a Bight = 85% of rope strength



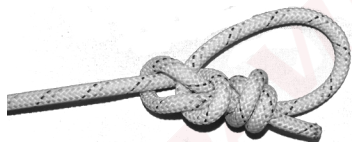
Alpine Butterfly = 65% of rope strength



Double Fisherman's = 60% of rope strength



Clove Hitch = 60% of rope strength



Bowline = 75% of rope strength